

The Science of Climate Change

Radiative balance and radiative forcing

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Department of Physics

Today's topics

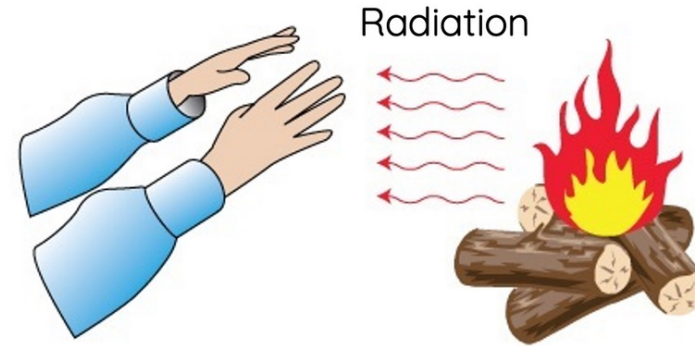
- Solar and terrestrial radiation
- The greenhouse effect
- Radiative forcing

Today's topics

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- The greenhouse effect
- Radiative forcing

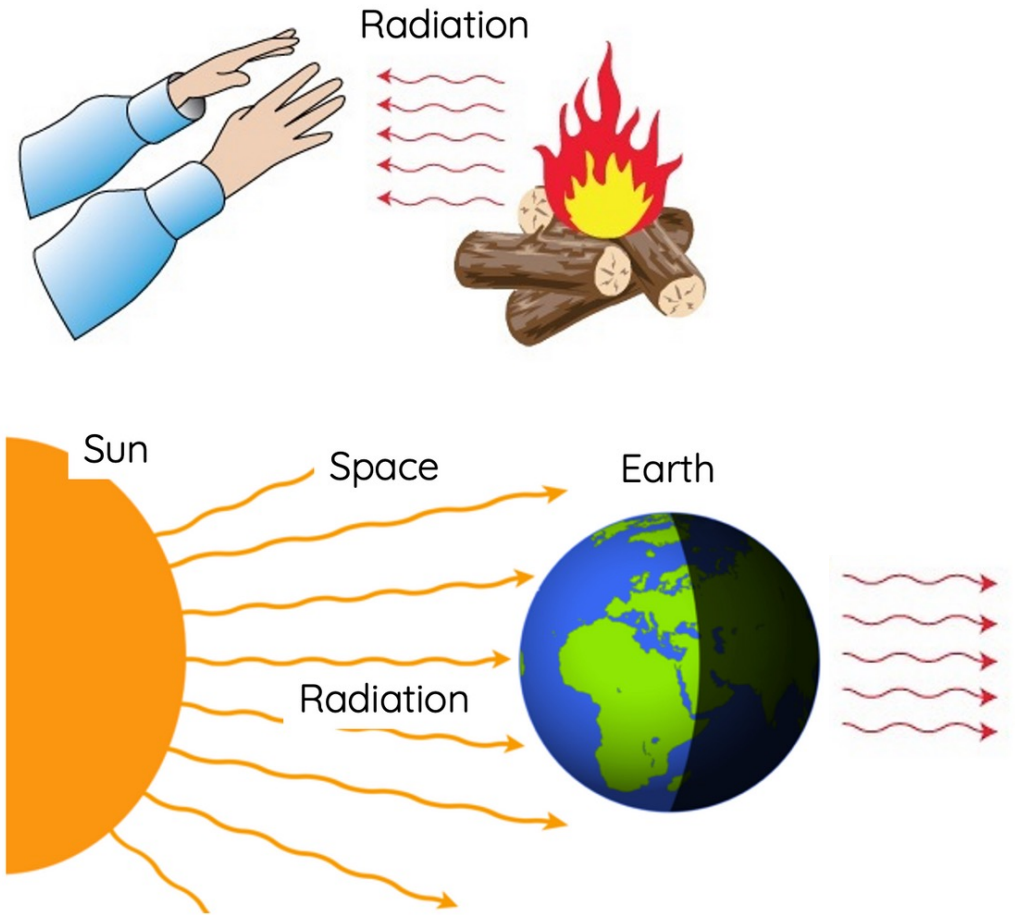
Radiation and heat

- **Radiation** is a form of transfer of **heat energy**

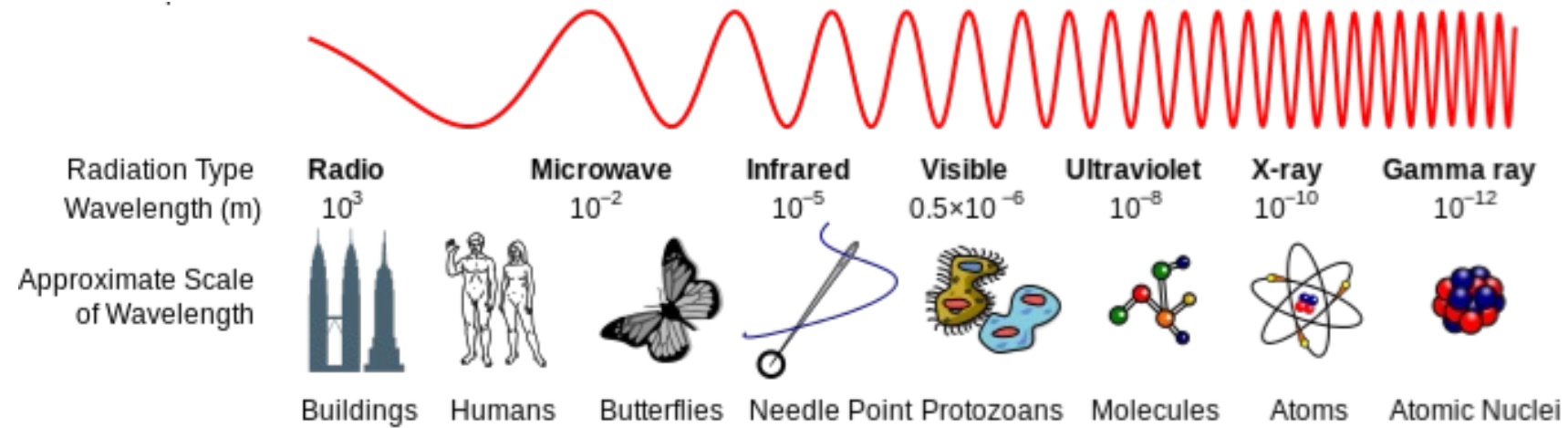


Radiation and heat

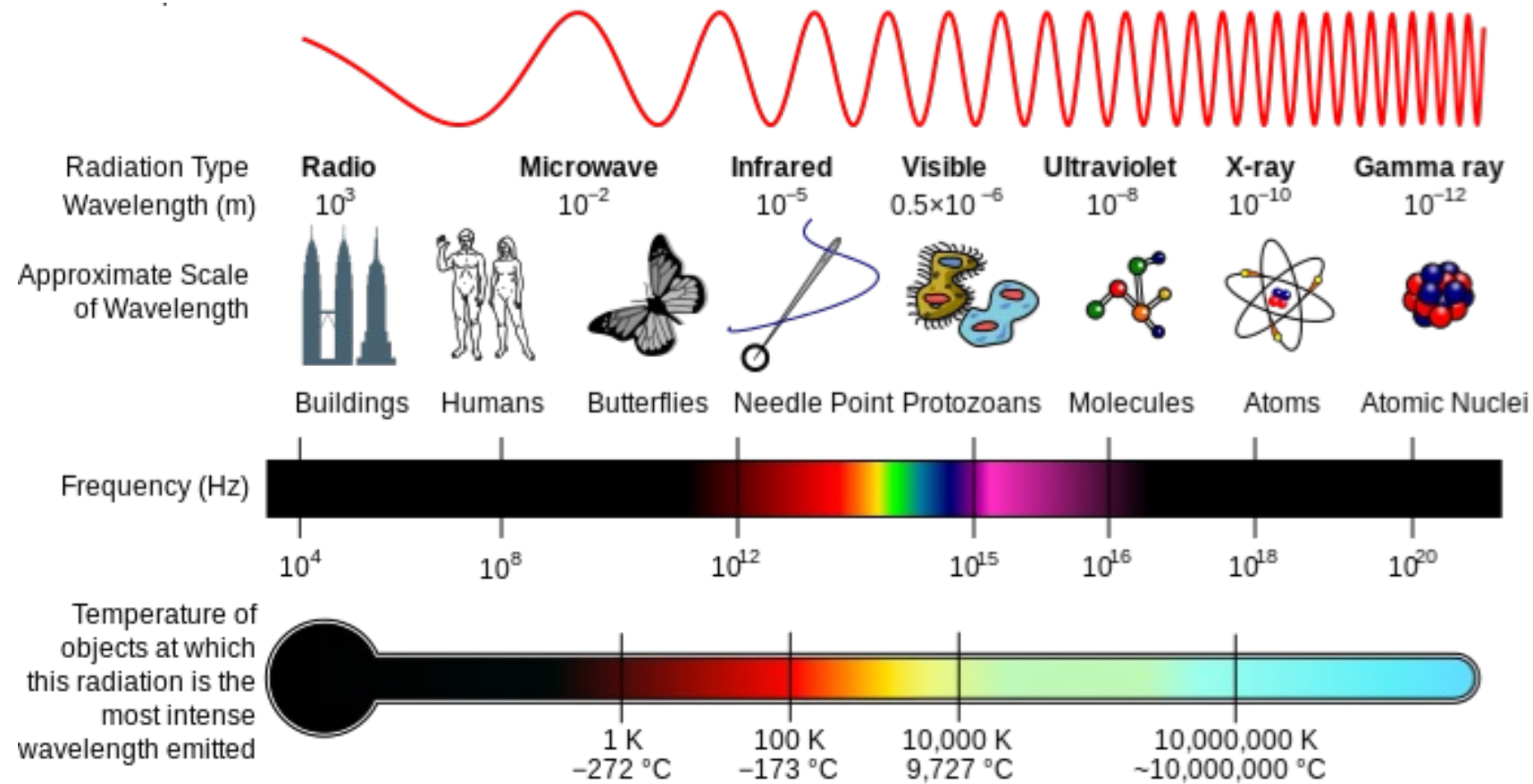
- **Radiation** is a form of transfer of **heat energy**
- The Sun's radiation heats the Earth
- Earth itself emits radiation to space



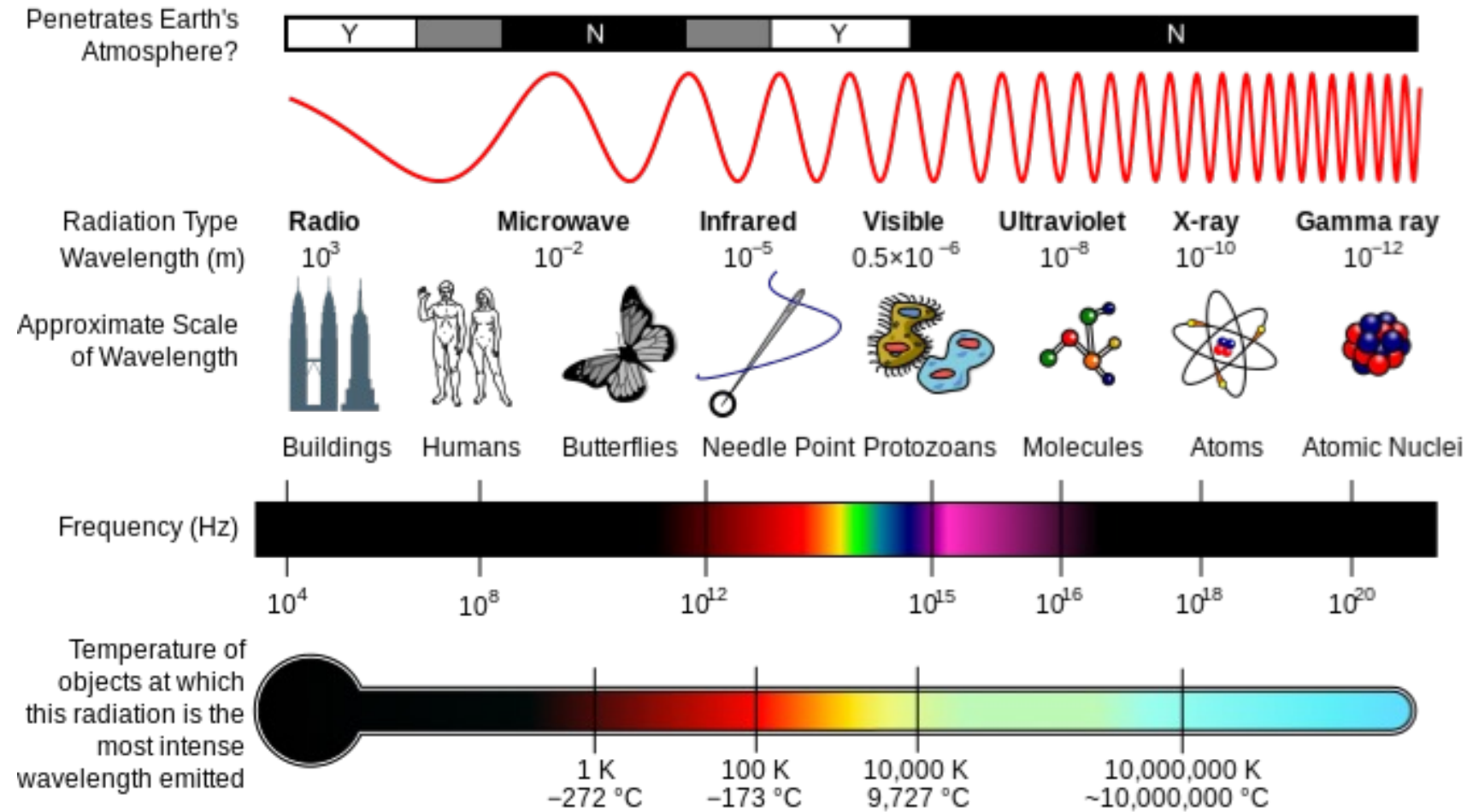
Electromagnetic radiation



Electromagnetic radiation

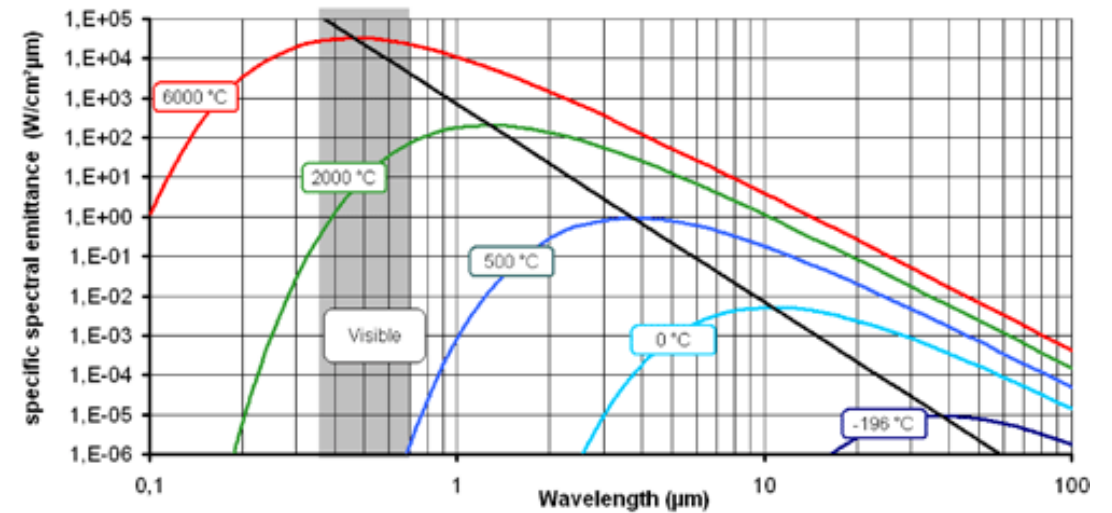


Electromagnetic radiation



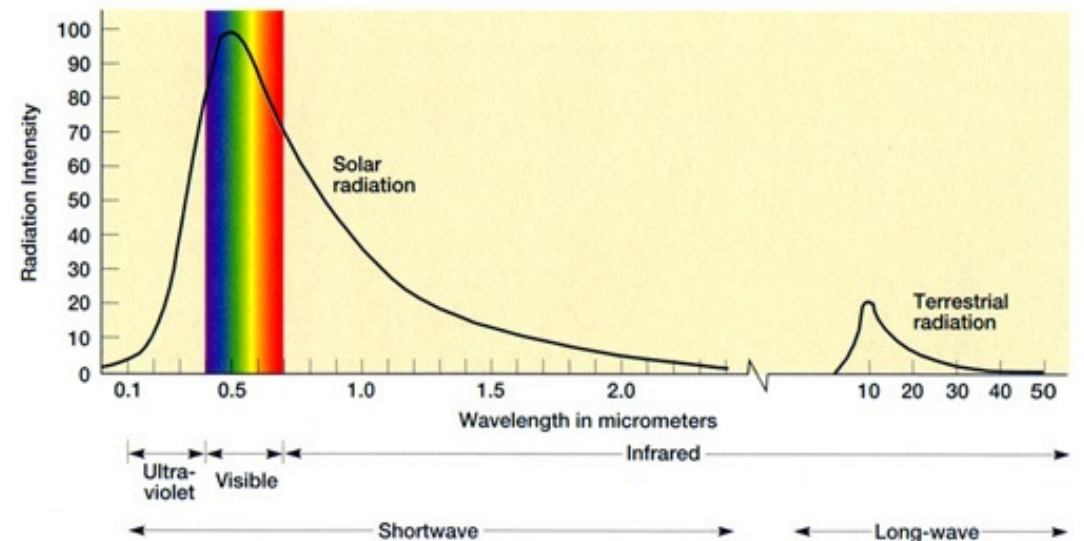
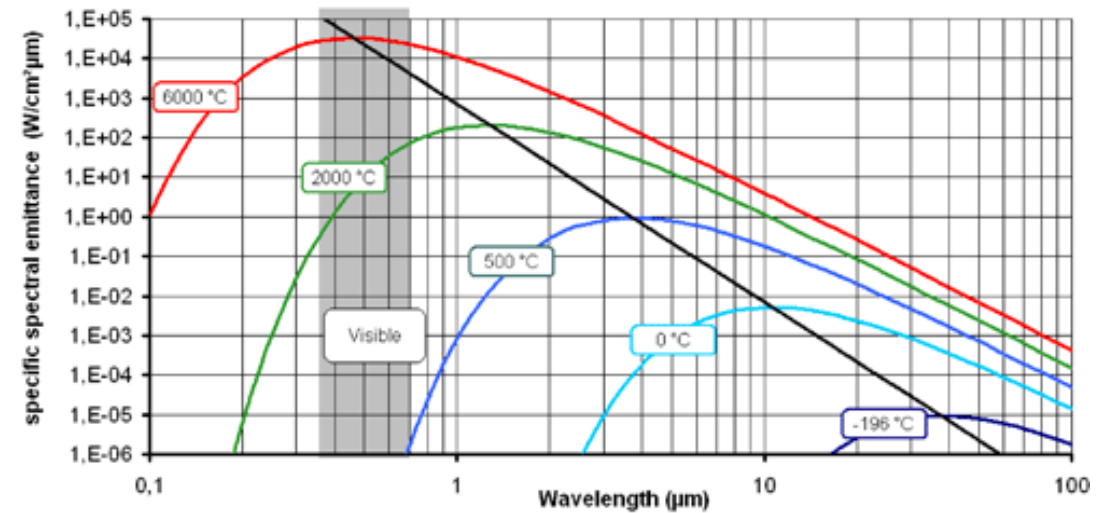
Planck's law

- Hotter objects emit more radiation (higher specific spectral emittance)
- The peak wavelength at which objects emit decreases with increasing temperature

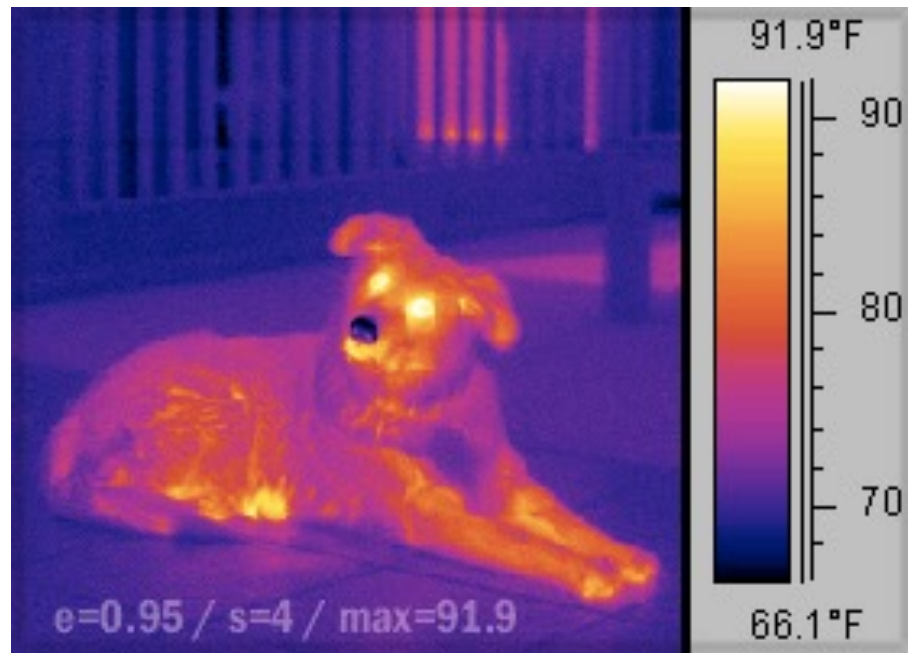


Planck's law

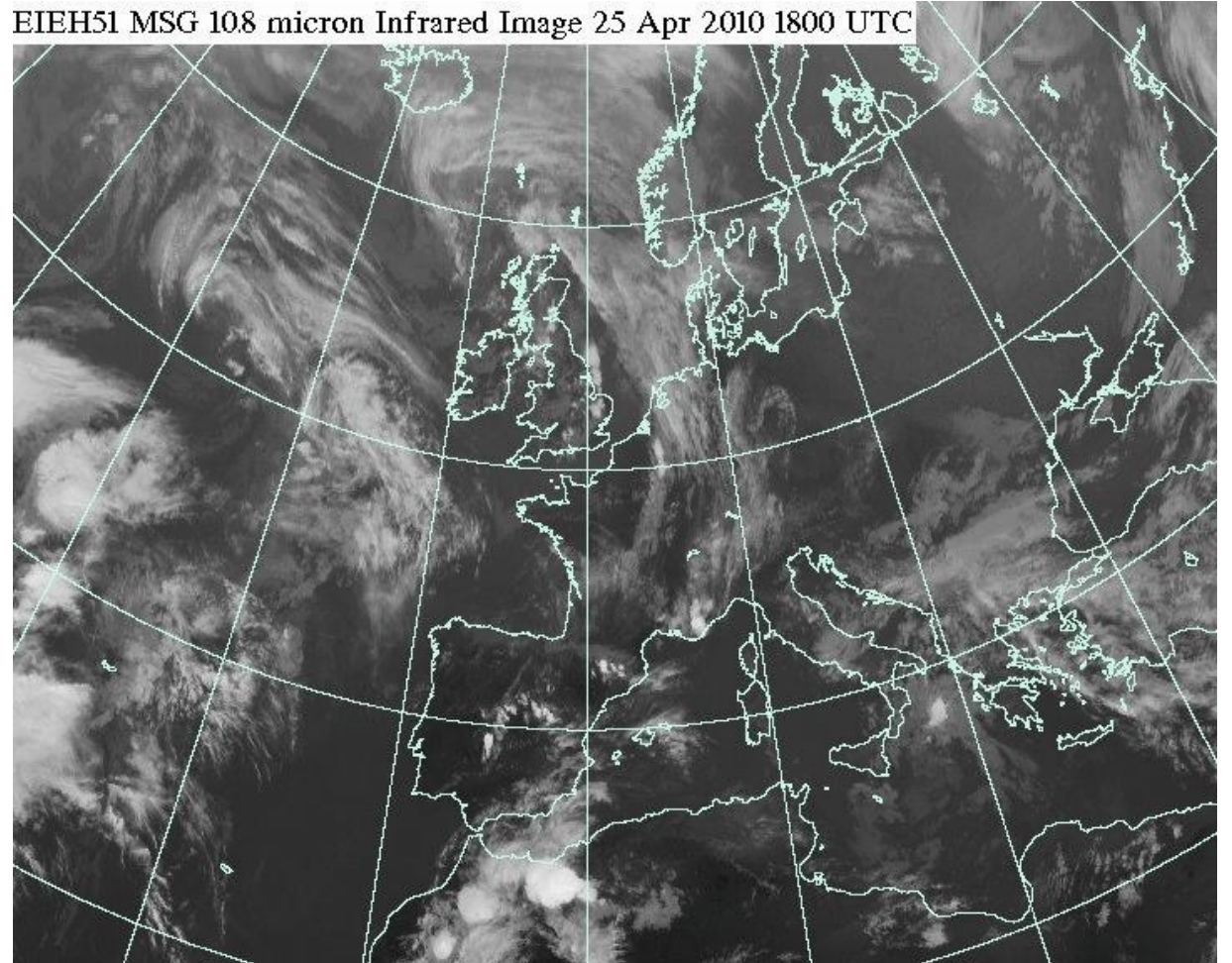
- Hotter objects emit more radiation (higher specific spectral emittance)
- The peak wavelength at which objects emit decreases with increasing temperature
- The sun's emission peaks in the visible part of the spectrum
- Earth, because it is so much colder, emits in the infrared part of the spectrum
 - Invisible to our eyes, but not to infrared cameras!



Infrared radiation examples

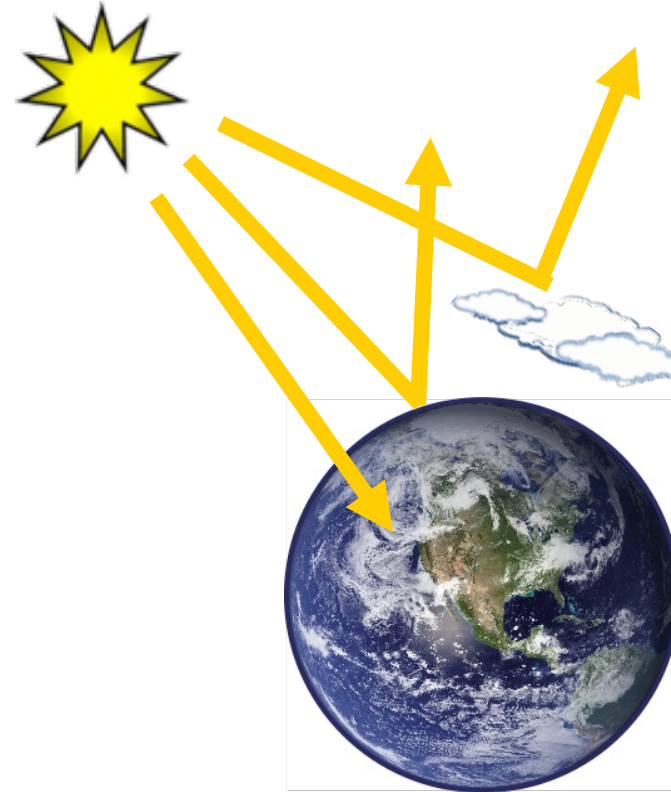


EIEH51 MSG 108 micron Infrared Image 25 Apr 2010 1800 UTC



The Earth's radiation balance

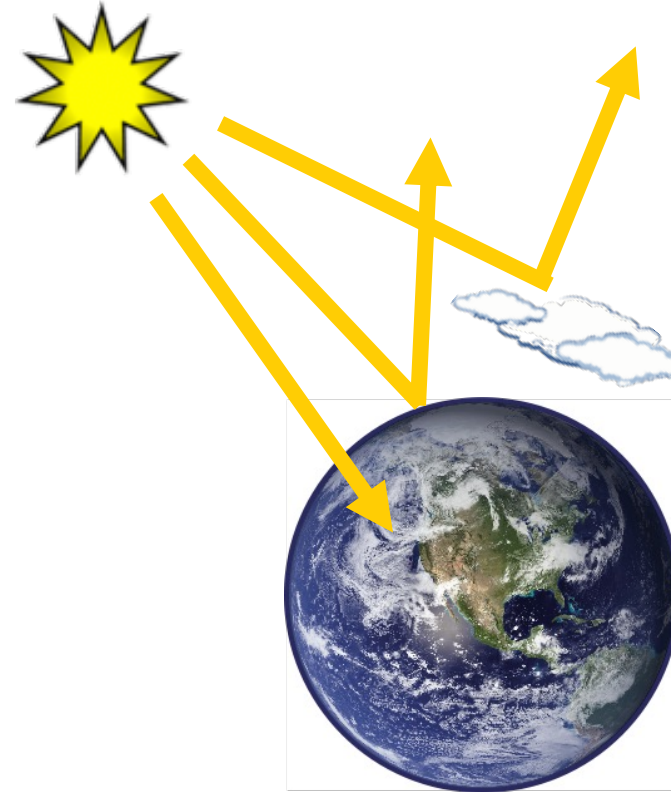
- All energy comes from the sun
- About 70% is absorbed by the Earth's surface and atmosphere
- 30% is reflected – especially by light surfaces like clouds, snow, deserts



(not to scale!)

The Earth's radiation balance

- All energy comes from the sun
- About 70% is absorbed by the Earth's surface and atmosphere
- 30% is reflected – especially by light surfaces like clouds, snow, deserts
- Radiative fluxes are measured in Watts per square meter, W/m^2
- The global-mean incoming solar radiation is $\sim 340 \text{ W/m}^2$
- So the net absorbed solar radiation is $70\% * 340 \text{ W/m}^2 \approx 240 \text{ W/m}^2$



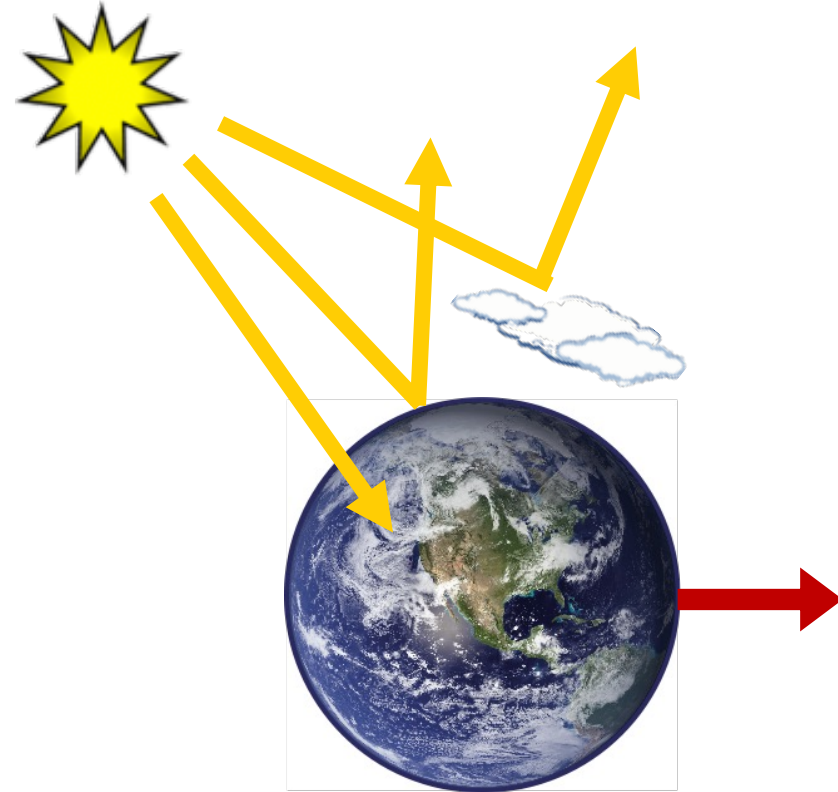
(not to scale!)

The Earth's radiation balance

- To achieve radiative balance, Earth must re-emit as much as it receives from the Sun:

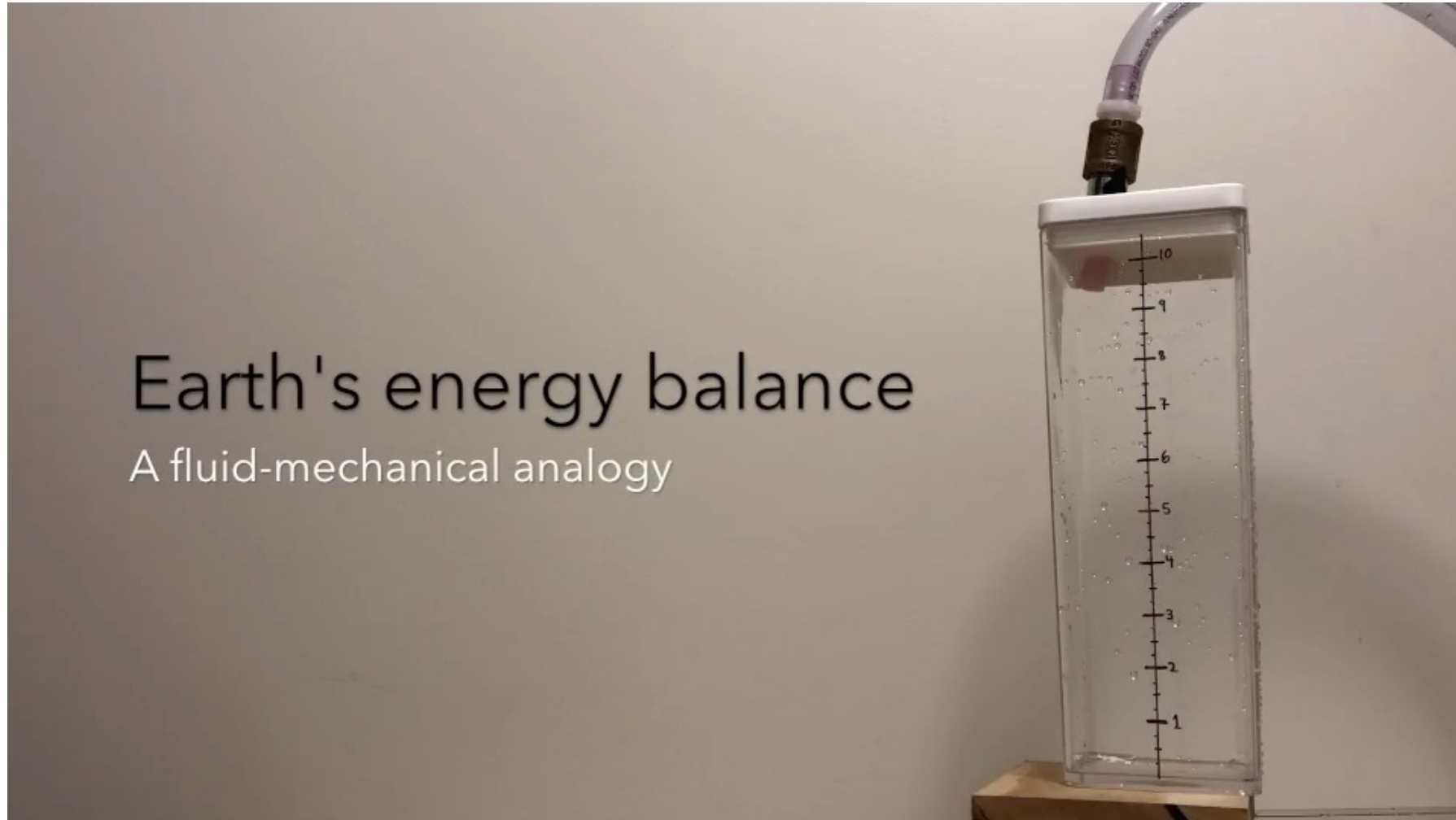
energy in = energy out

- It does so by emitting **infrared radiation** to space



(not to scale!)

The Earth's radiation balance

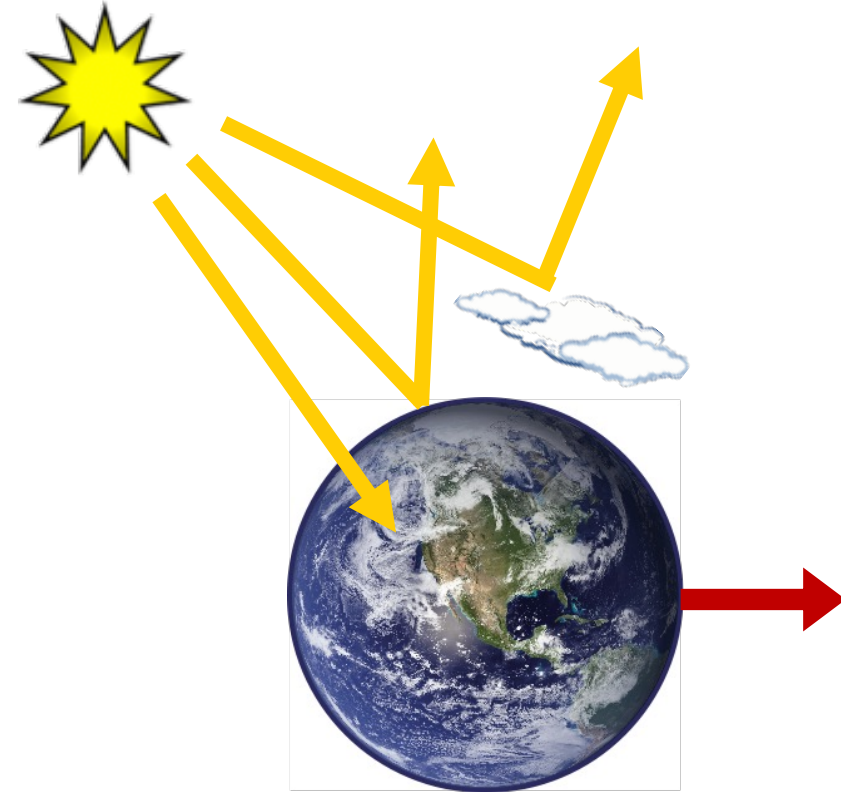


The Earth's radiation balance

- To achieve radiative balance, Earth must re-emit as much as it receives from the Sun:

energy in = energy out

- It does so by emitting **infrared radiation** to space
- Since the emission of radiative energy scales with temperature, can calculate the equilibrium surface temperature to satisfy energy balance (lecture 7)
 - This yields -19°C (!)
 - What's missing?!



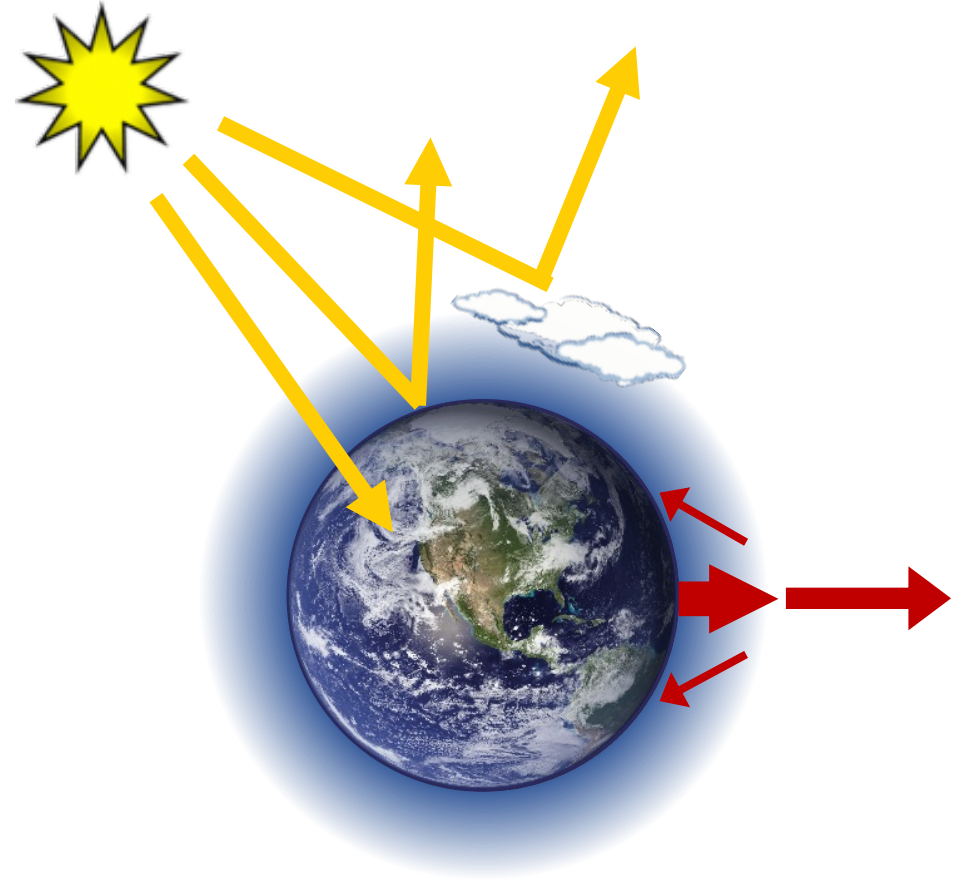
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Today's topics

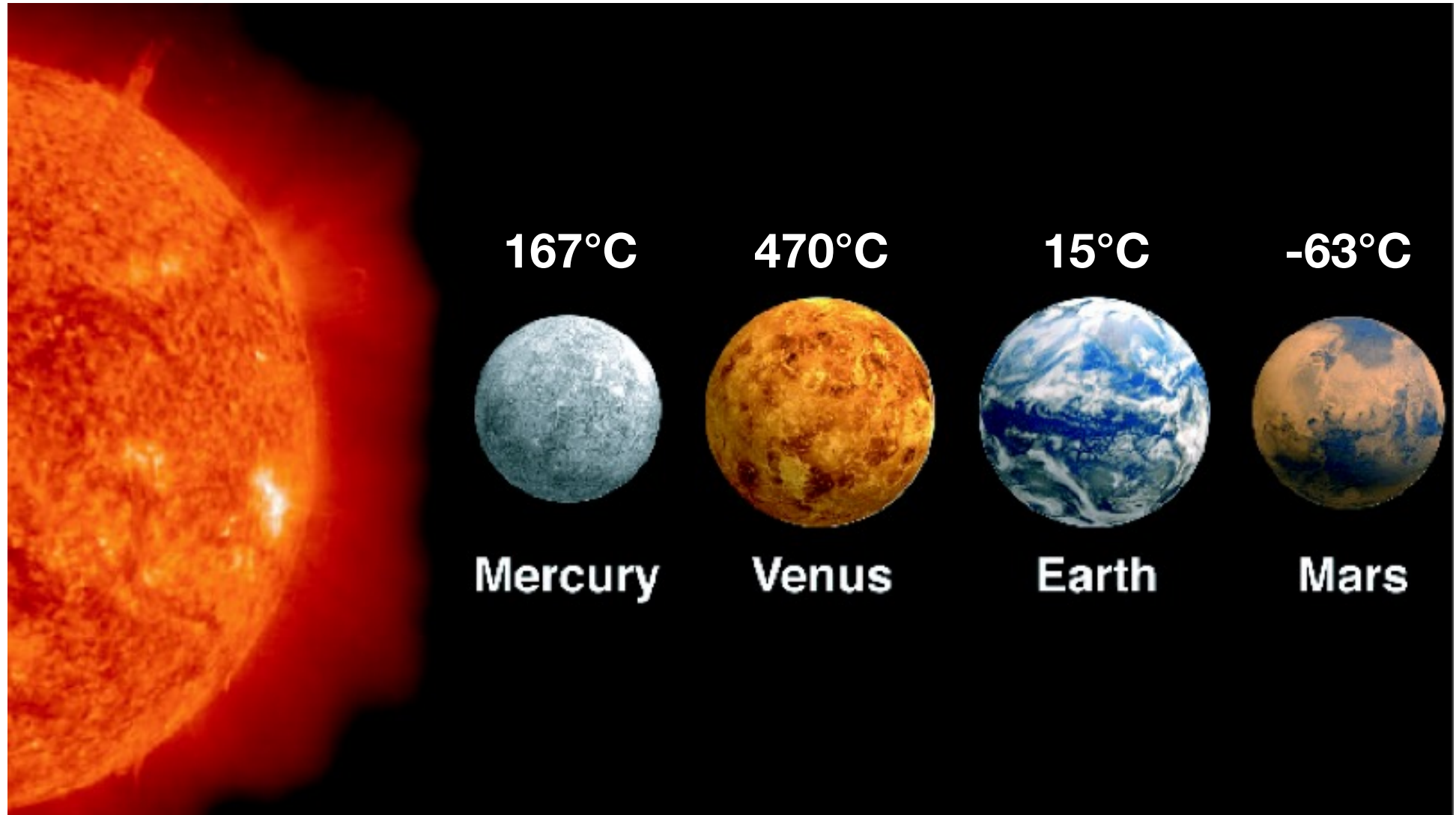
- Solar and terrestrial radiation
- The greenhouse effect
- Radiative forcing

The greenhouse effect

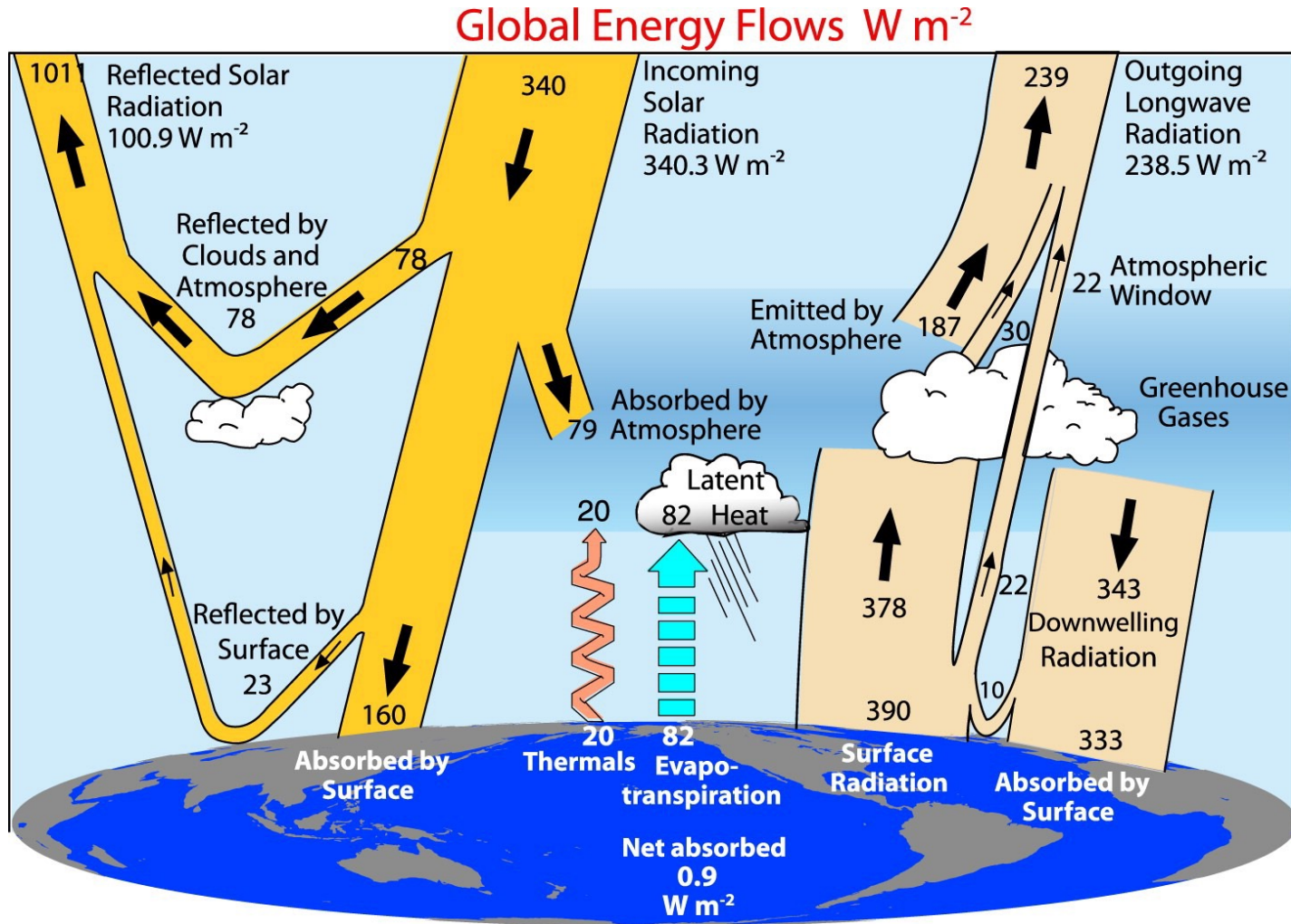
- Some of the radiation emitted by Earth is trapped by gases in the atmosphere and re-emitted in all directions
- The fact that some of the energy is re-emitted downward back to the surface is the **greenhouse effect**
- The average surface temperature with the greenhouse effect is 15°C
- So, in principle, the greenhouse effect is a good thing!



The greenhouse effect on other planets

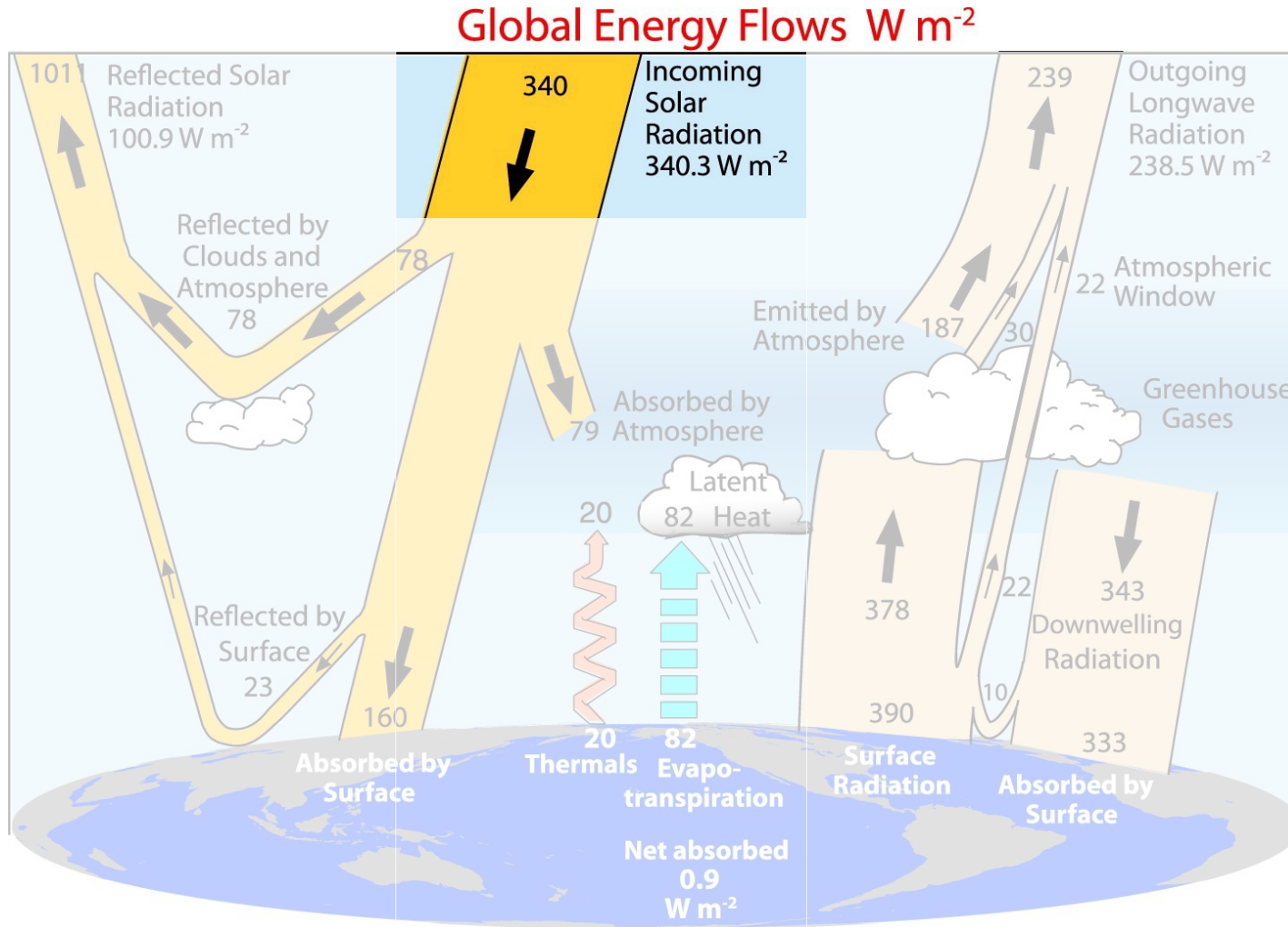


A detailed look at the radiative budget

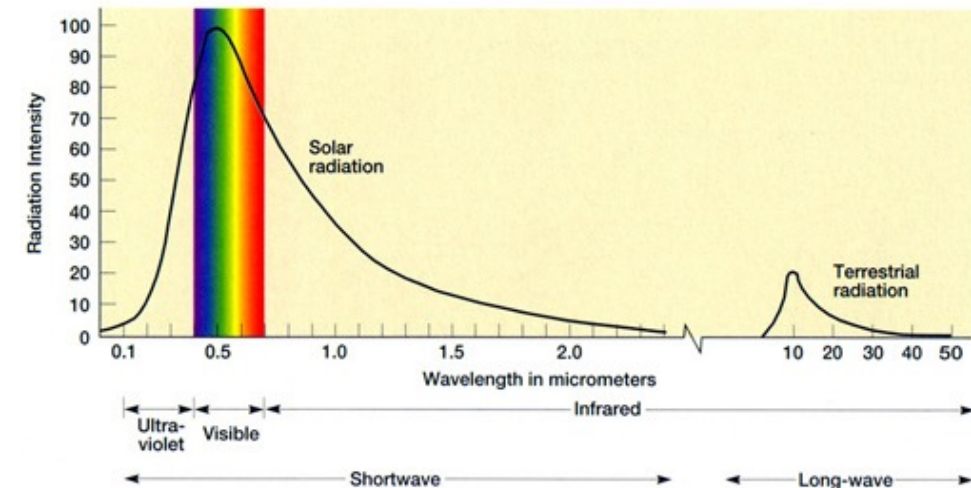


From Trenberth 2020

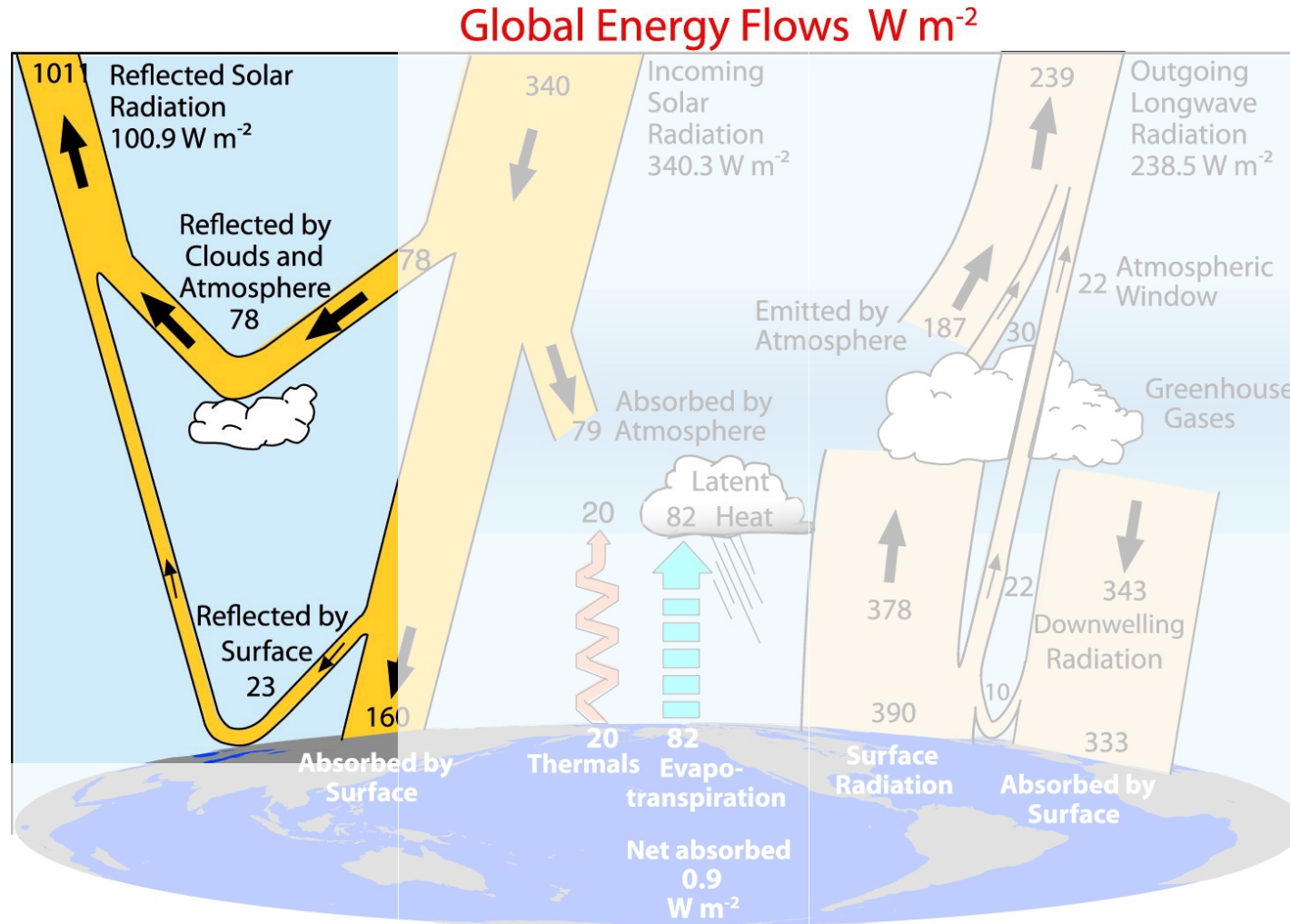
Shortwave radiation



- The Sun provides all of the energy
- Solar radiation is referred to as **shortwave** radiation, because it's a short wavelength compared to Earth's infrared radiation



Shortwave radiation

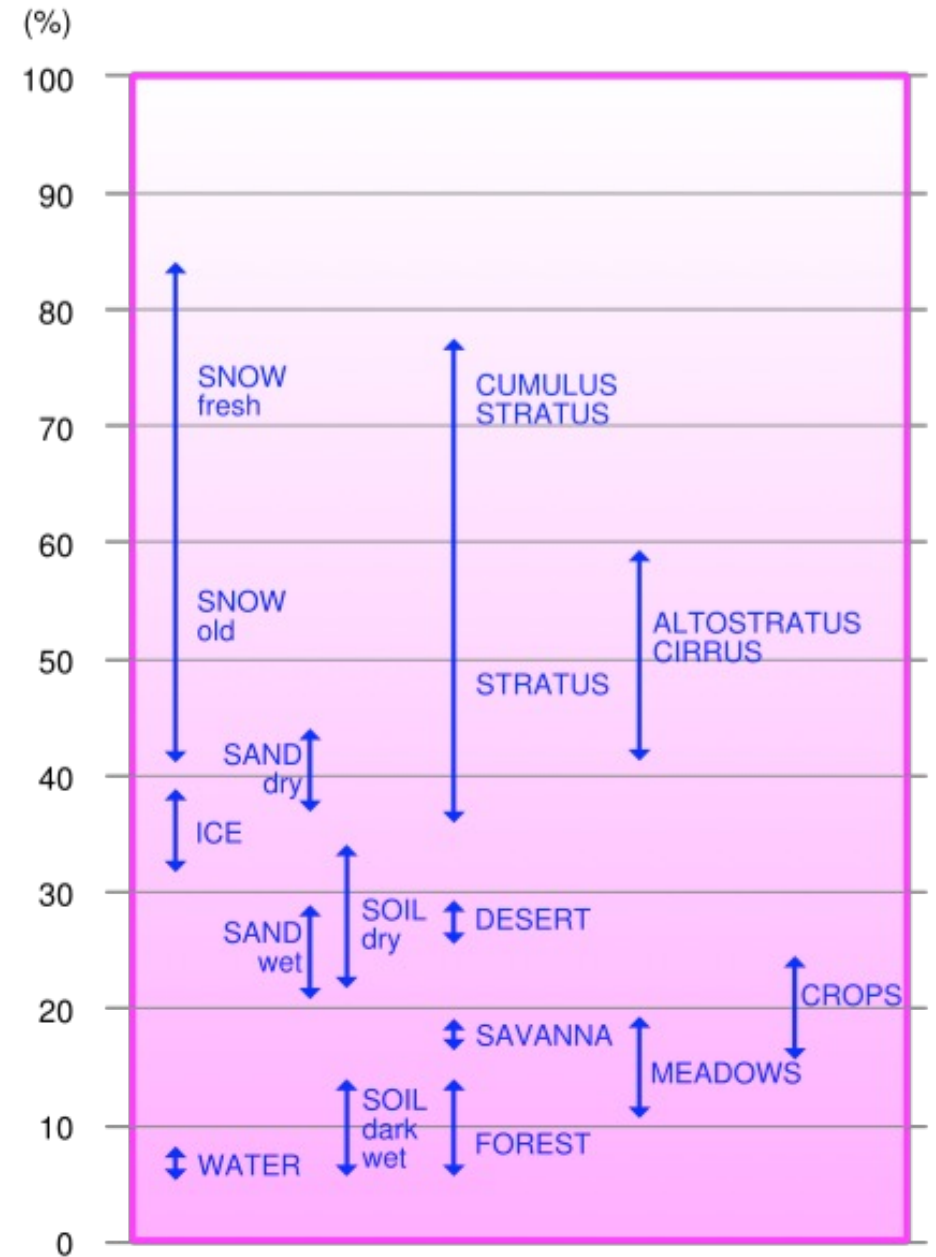


- The reflection of solar energy depends on the **albedo** of the Earth's atmosphere and surface
- Albedo = (reflected solar radiation) / (incoming solar radiation), a number between 0 and 1
- Light, reflective surfaces (snow, clouds) tend to have high albedos; dark surfaces (water, forests) have low albedos

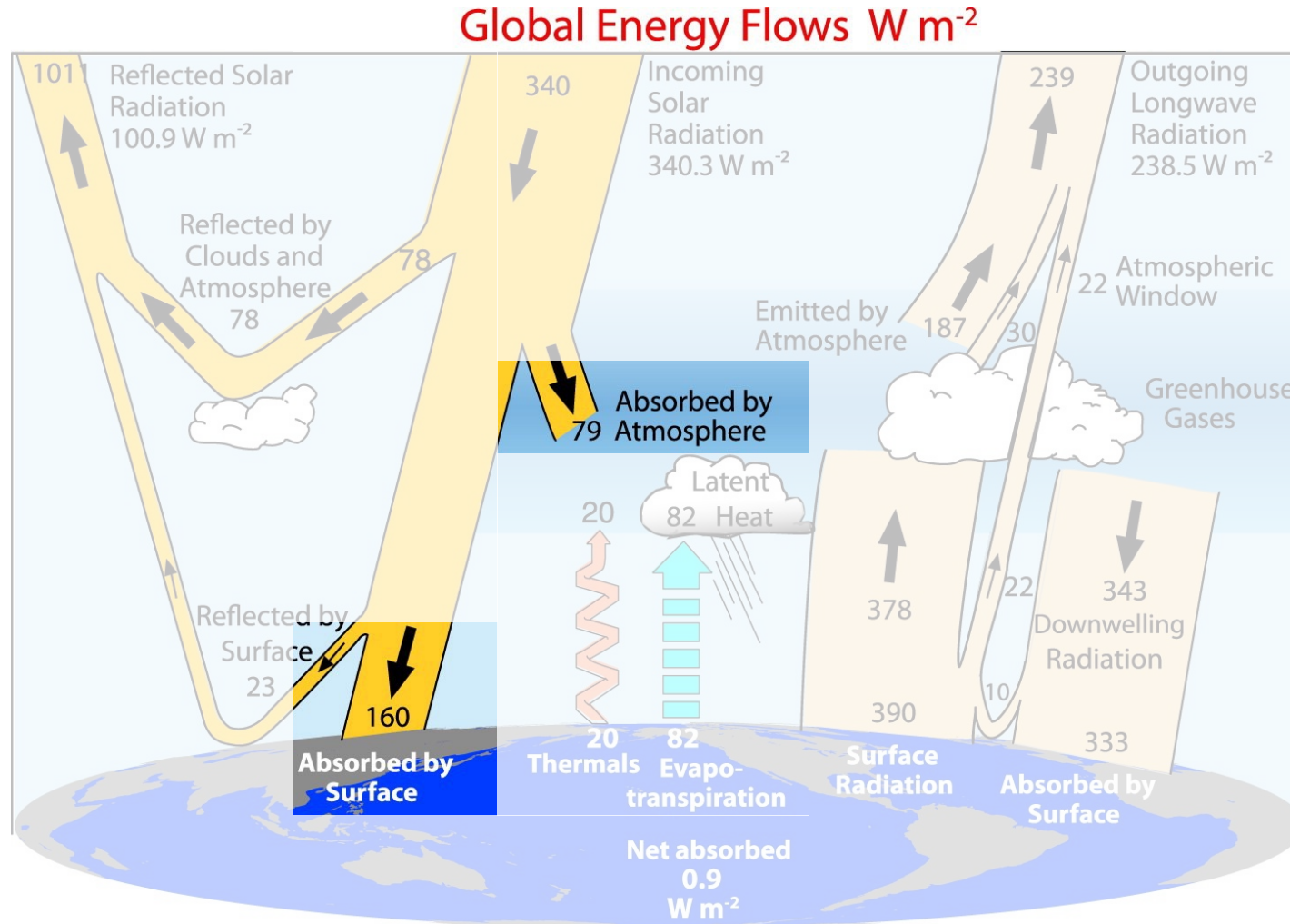
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Albedo

- The Earth's mean albedo is 30%
- However, the albedo strongly varies spatially, depending on surface type



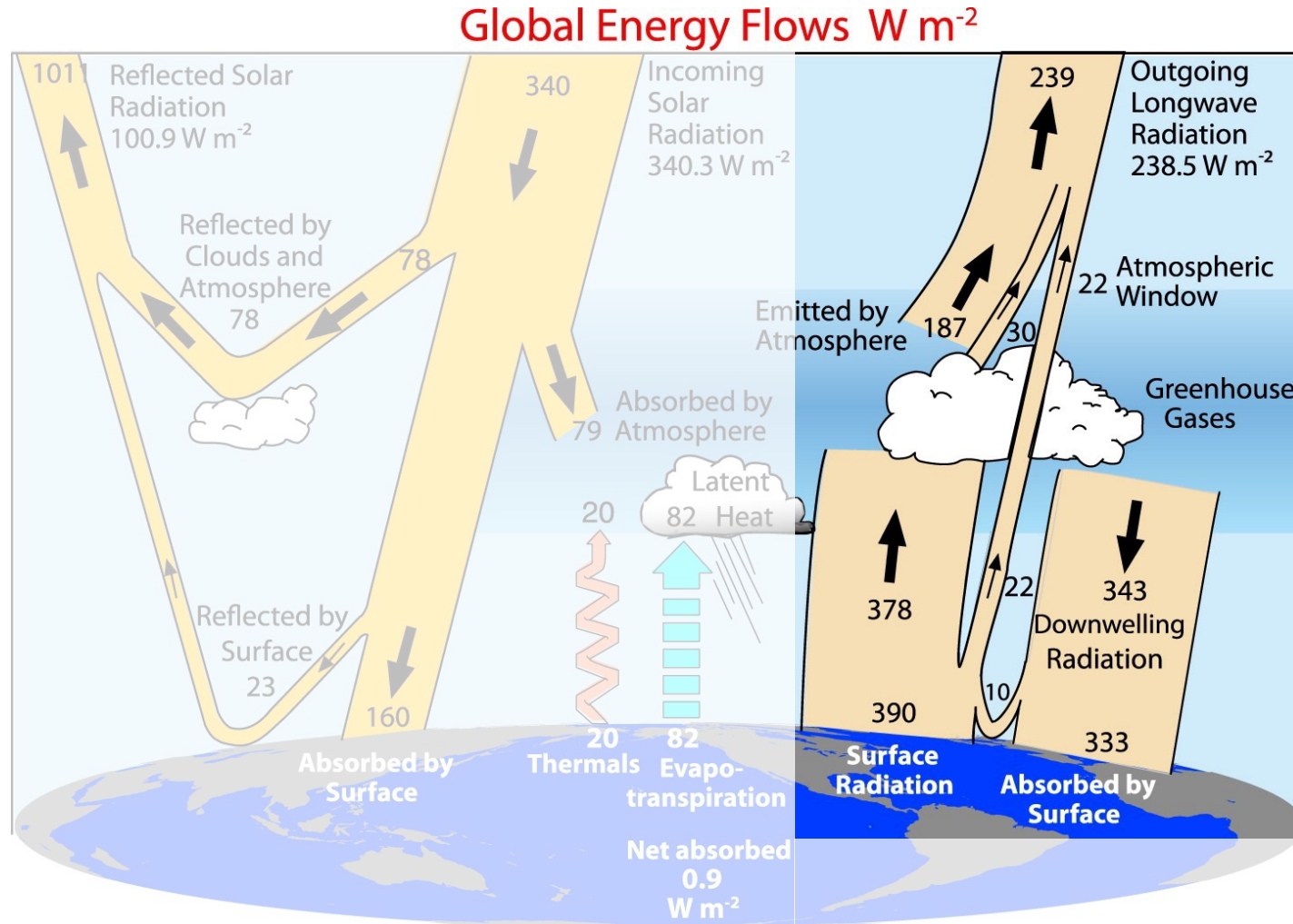
Shortwave radiation



- What is not reflected is absorbed (about 70%)
- Most of it is absorbed at the surface
- However, atmospheric gases also absorb solar radiation (e.g. ozone, water vapour)

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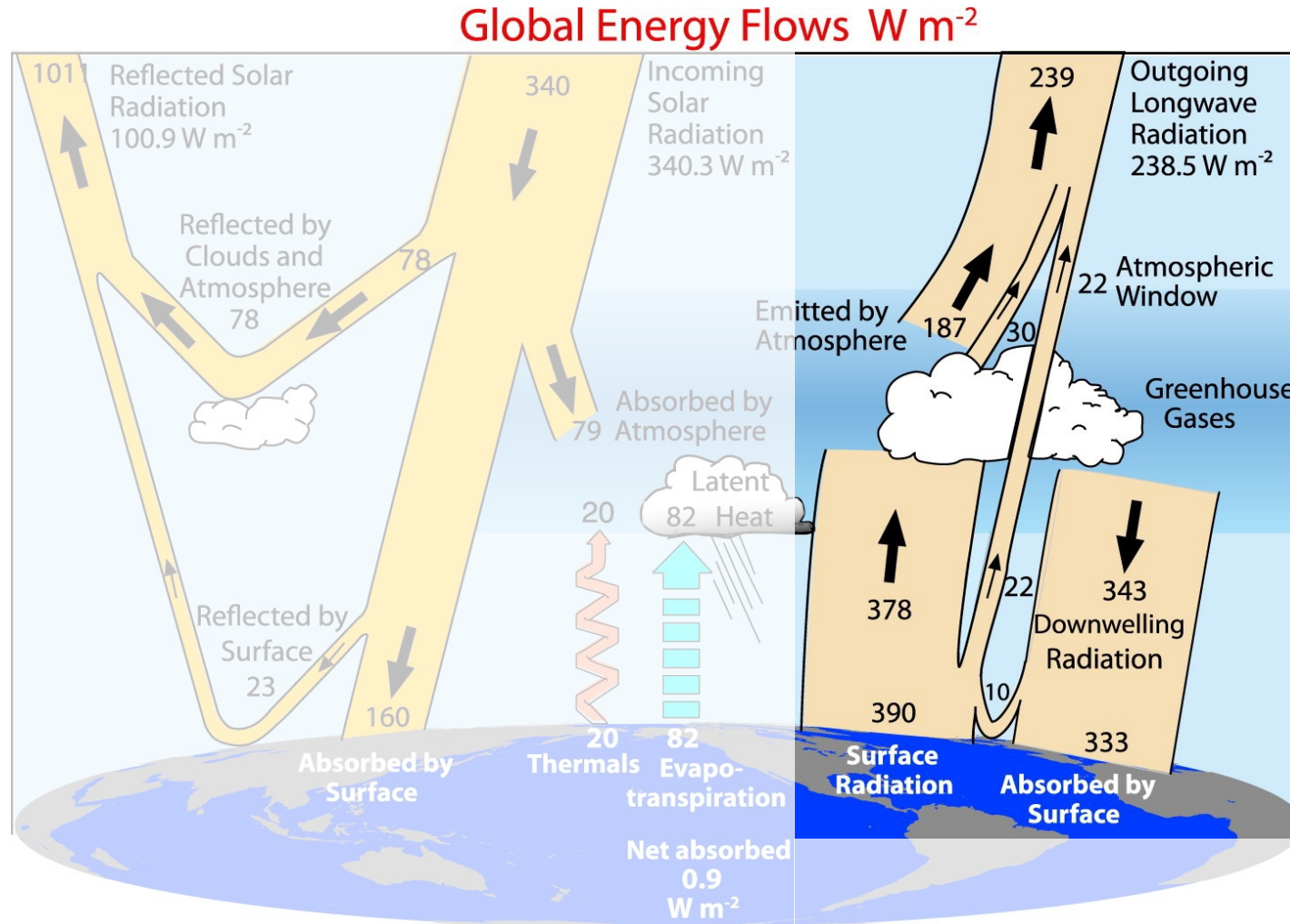
Longwave radiation



- Now let's look at terrestrial radiation
- Terrestrial infrared radiation is commonly referred to as **longwave**

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Longwave radiation

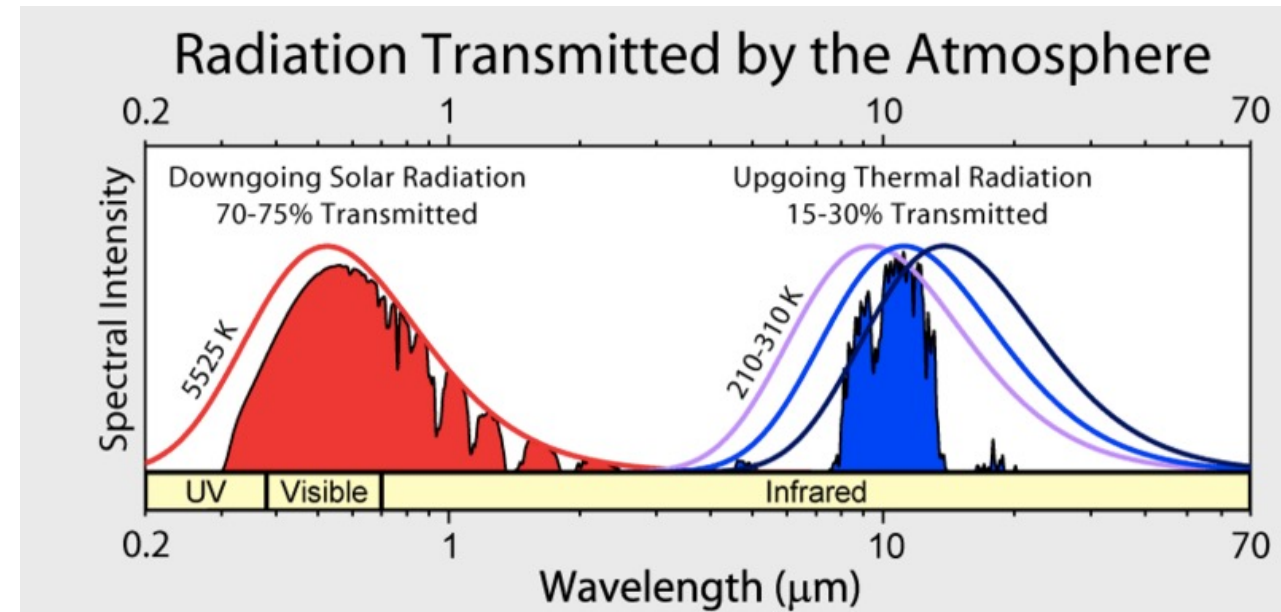


- Large up and down longwave fluxes: this is the greenhouse effect
- Most of the longwave radiation emitted from the surface and lower atmosphere is absorbed, then re-emitted in all directions

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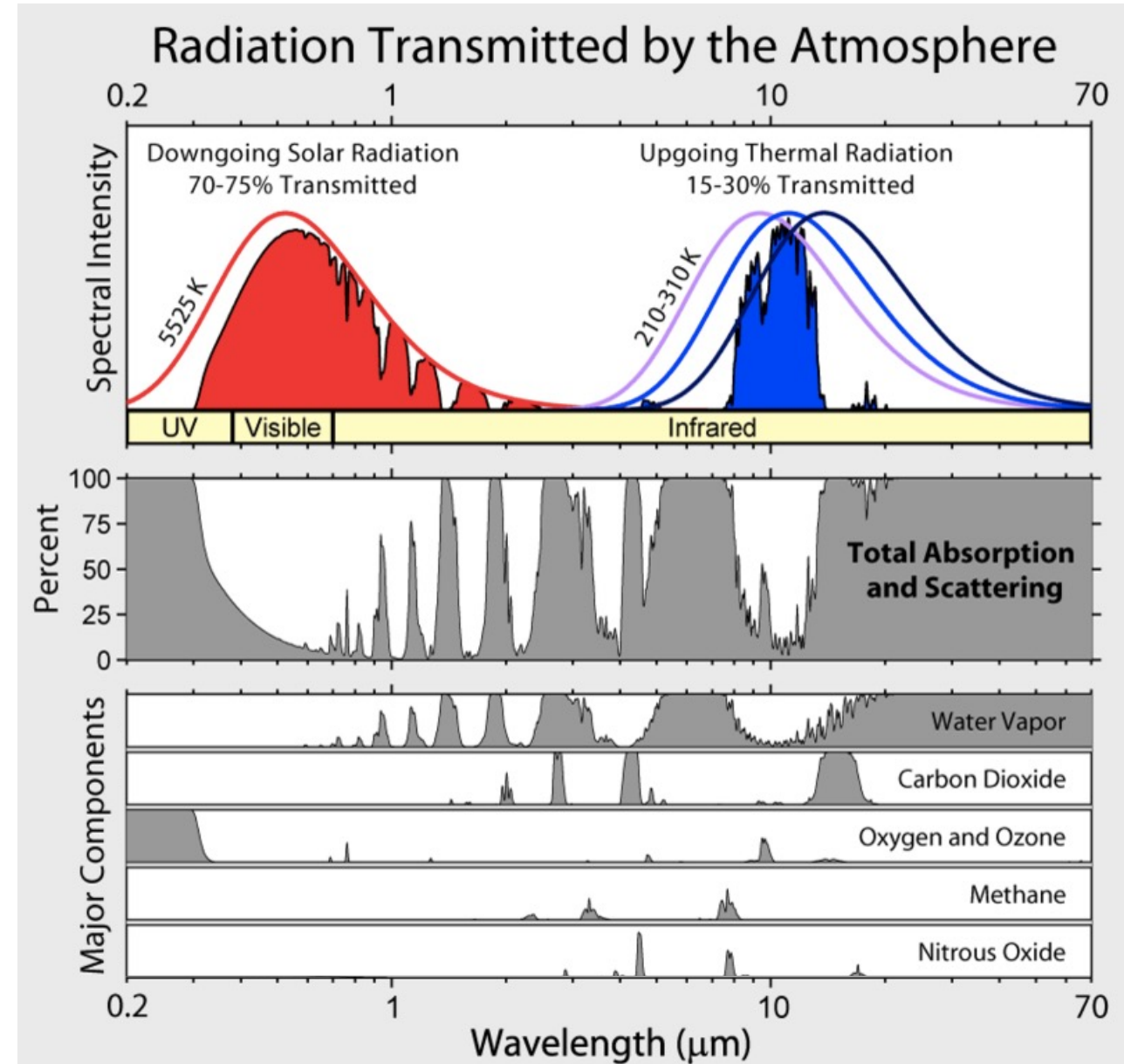
Greenhouse gases

- Solar radiation is short wavelength (shortwave)
- Terrestrial radiation is longer wavelength (longwave)
- Most solar radiation passes through the atmosphere
- Only a fraction of terrestrial radiation passes directly through the atmosphere



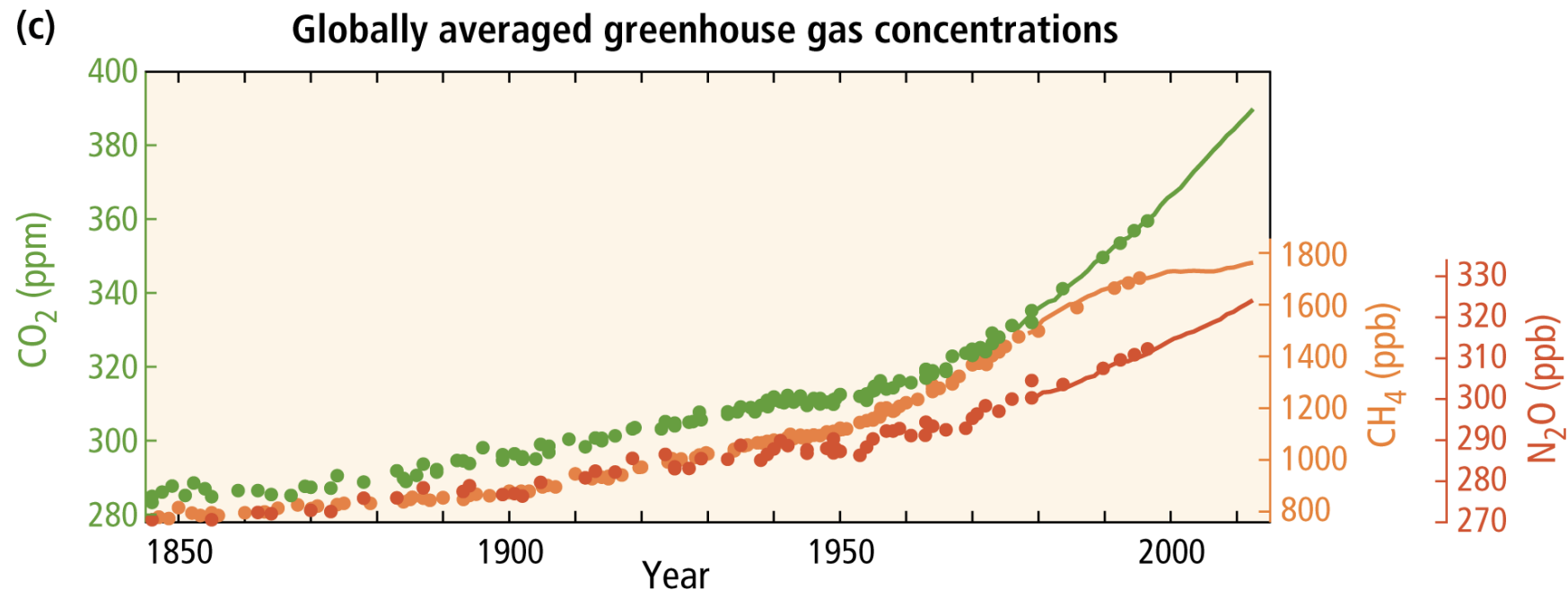
Greenhouse gases

- There are many greenhouse gases – CO₂ is just one of them
- Main GHG is **water vapour**, opaque to infrared radiation at many wavelengths
- Additional important GHGs are ozone (O₃), methane (CH₄), nitrous oxide (N₂O), and chlorofluorocarbons (CFCs), but there are more
- Most GHGs exist naturally, but they are perturbed owing to human activities

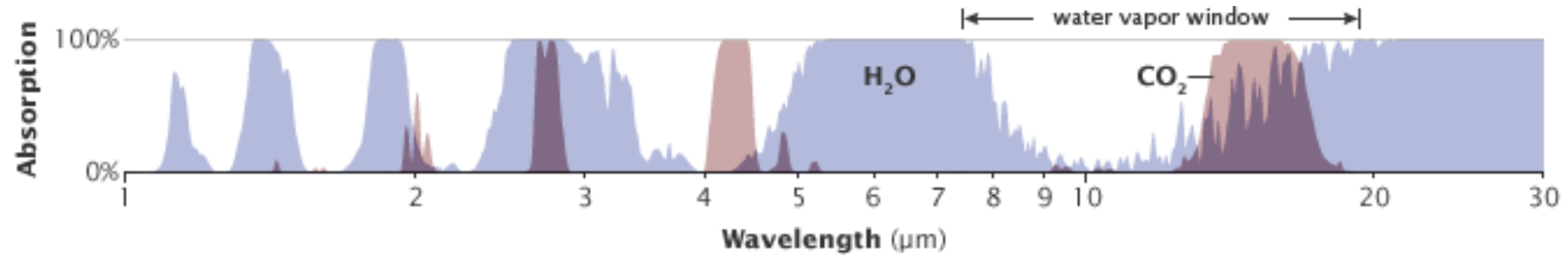


Greenhouse gases

- CH_4 has more than doubled, N_2O also rapidly increasing
- CH_4 from natural gas leaks, livestock
- N_2O from industrial processes, cars, fertilisers

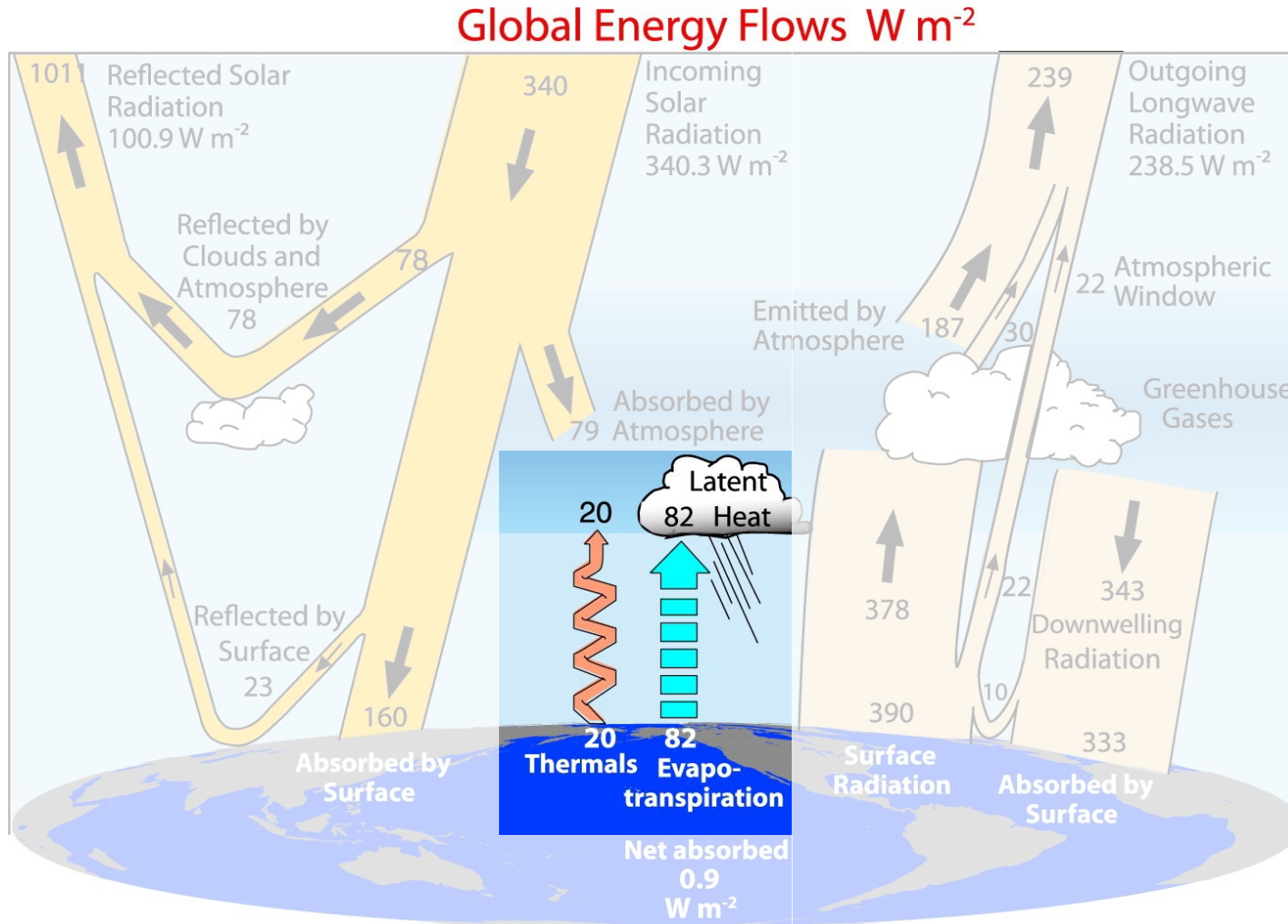


The water vapour window



- The absorption of infrared radiation by greenhouse gases strongly depends on wavelength (this itself is related to the molecular structure of the gases)
- CO₂ is important because it partially masks the “water vapour window”, a range of wavelengths where the Earth is able to emit to space
- Increasing CO₂ concentrations partially “close” the window

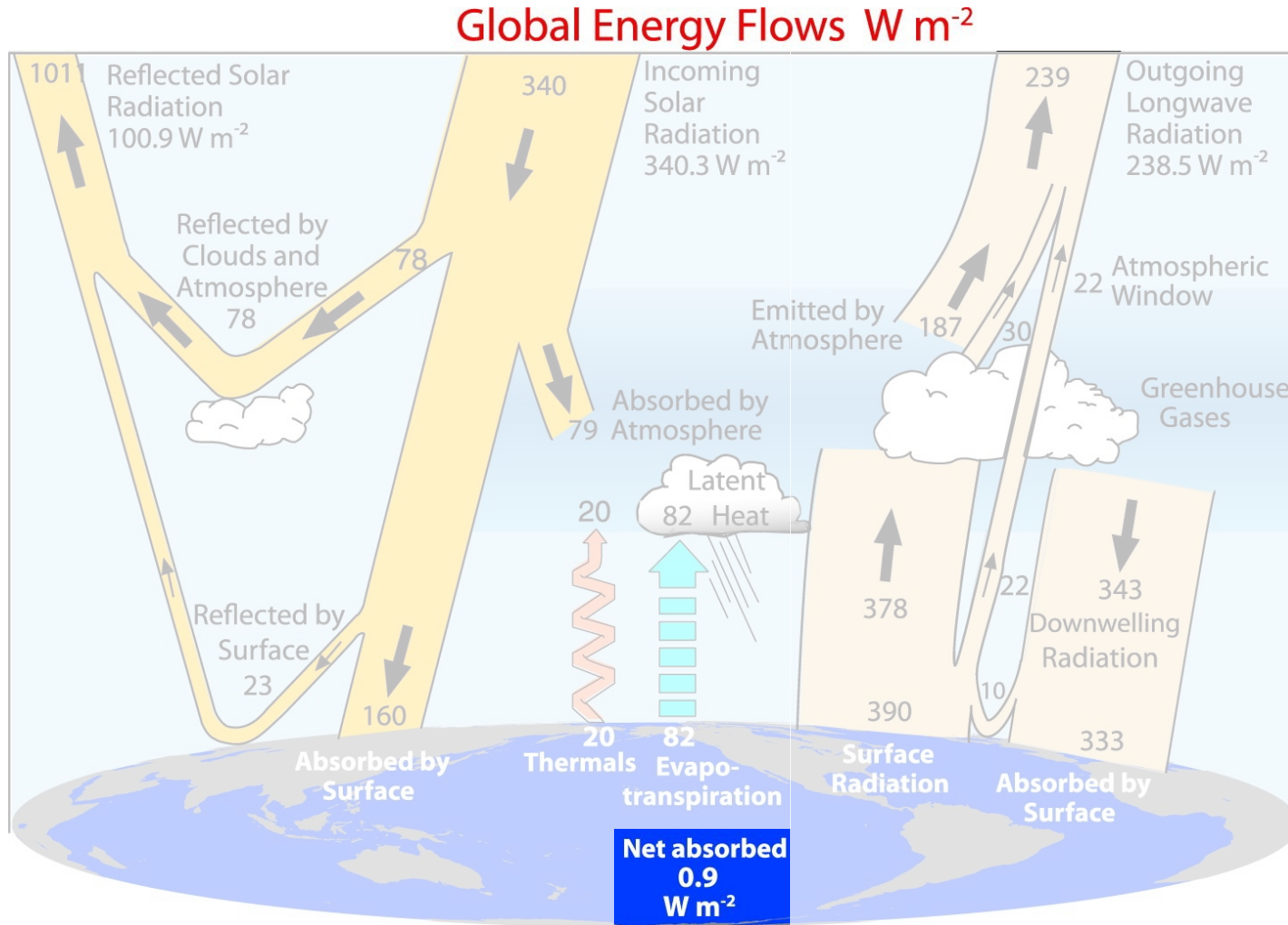
Sensible and latent heat fluxes



- Some of the energy exchange between surface and atmosphere is via fluxes of sensible heat (“temperature”) and latent heat (humidity)
- This is not radiation! (physically, it’s conduction and convection)

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Radiative imbalance



- Because the Earth is warming, there must be a positive imbalance: Earth is absorbing more than it emits
- This imbalance is mainly due to the strengthening of the greenhouse effect (though there are additional feedbacks – see later)

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Today's topics

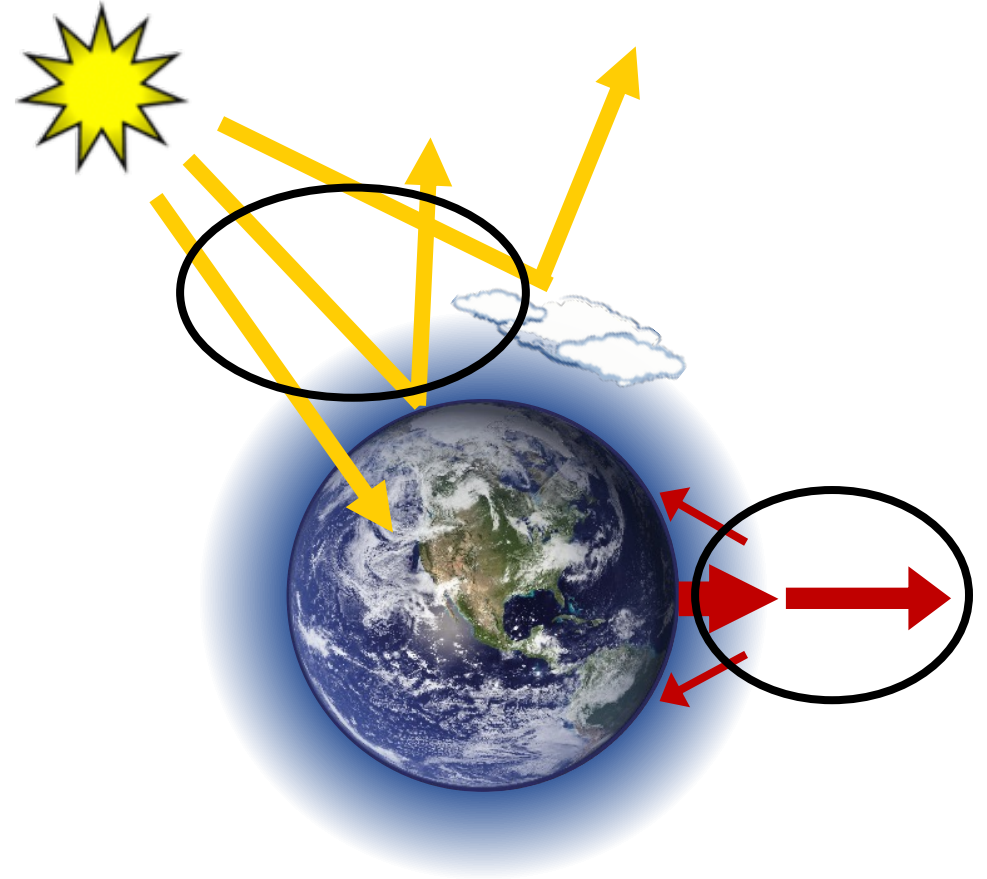
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Perturbing radiative balance: radiative forcing

- Radiative forcing is an externally imposed perturbation to the radiative balance
- Measured in W/m^2 , relative to some reference state (usually pre-industrial)

Perturbing radiative balance: radiative forcing

- Radiative forcing is an externally imposed perturbation to the radiative balance
- Measured in W/m^2 , relative to some reference state (usually pre-industrial)
- Can force either by perturbing “**energy in**” (how much solar radiation is absorbed) or “**energy out**” (how much terrestrial radiation escapes to space)

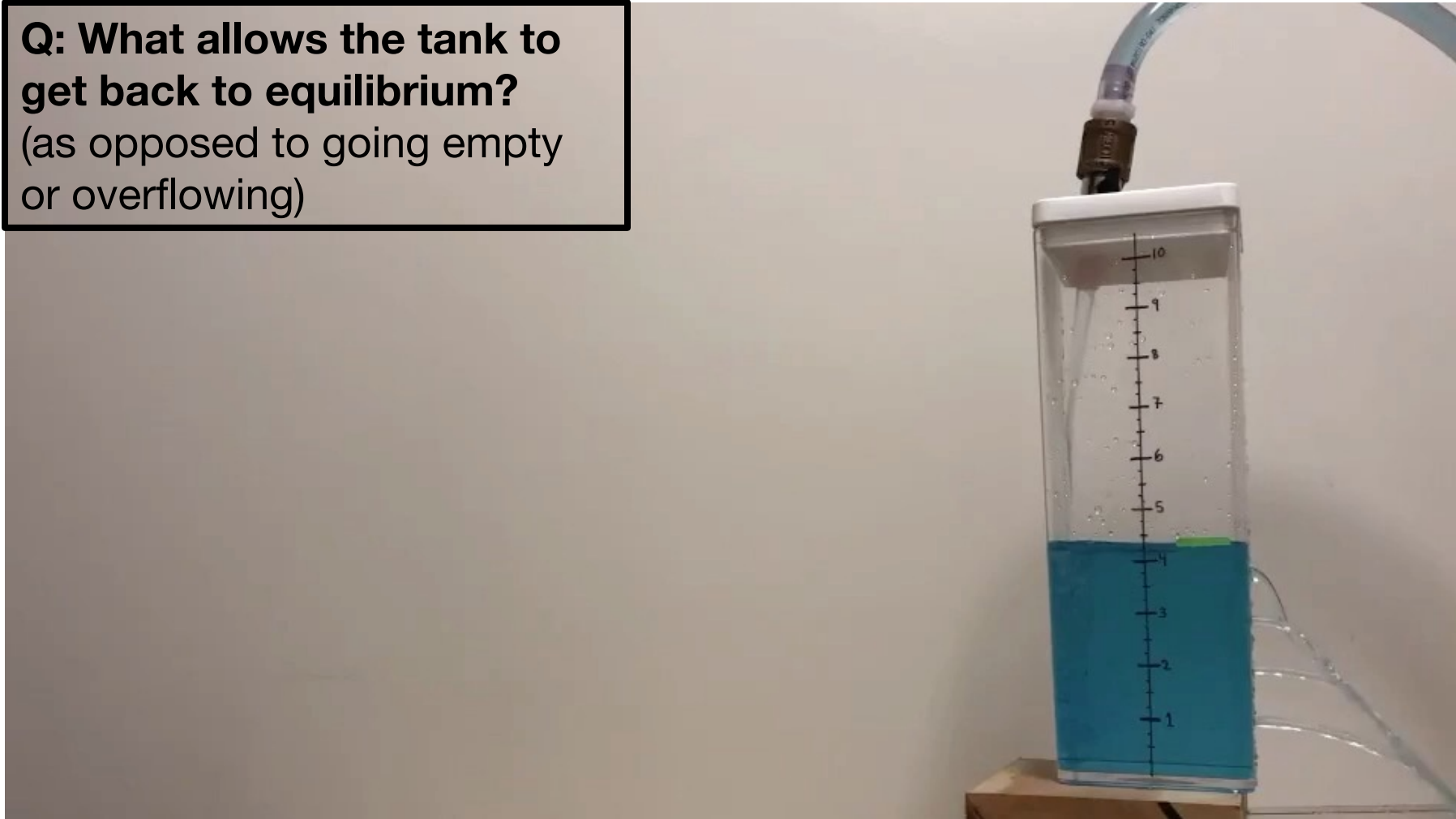


Perturbing radiative balance: radiative forcing



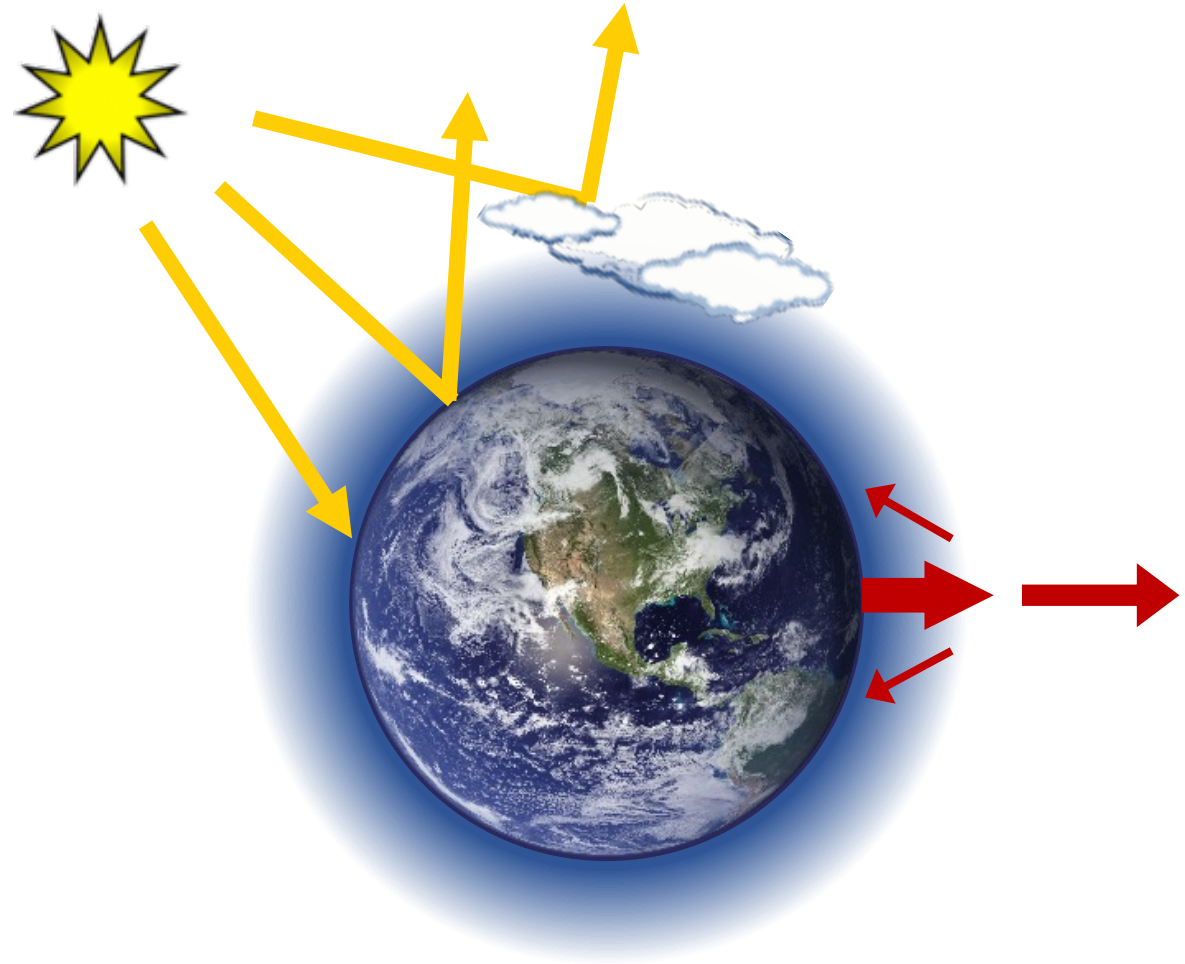
Perturbing radiative balance: radiative forcing

Q: What allows the tank to get back to equilibrium?
(as opposed to going empty or overflowing)



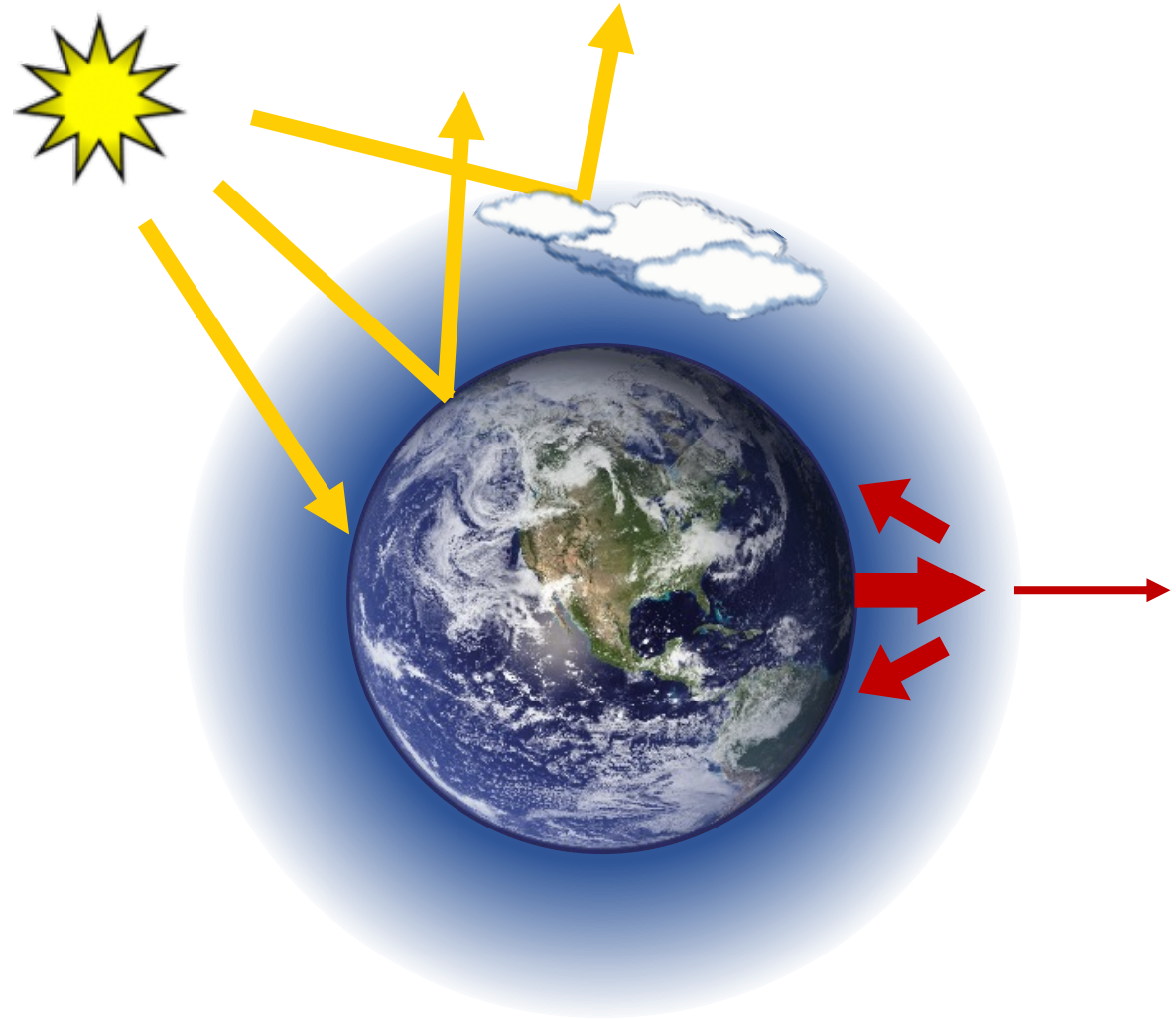
Perturbing radiative balance: GHG forcing

- Let's consider the case of GHG forcing
- At equilibrium:
energy in = energy out



Perturbing radiative balance: GHG forcing

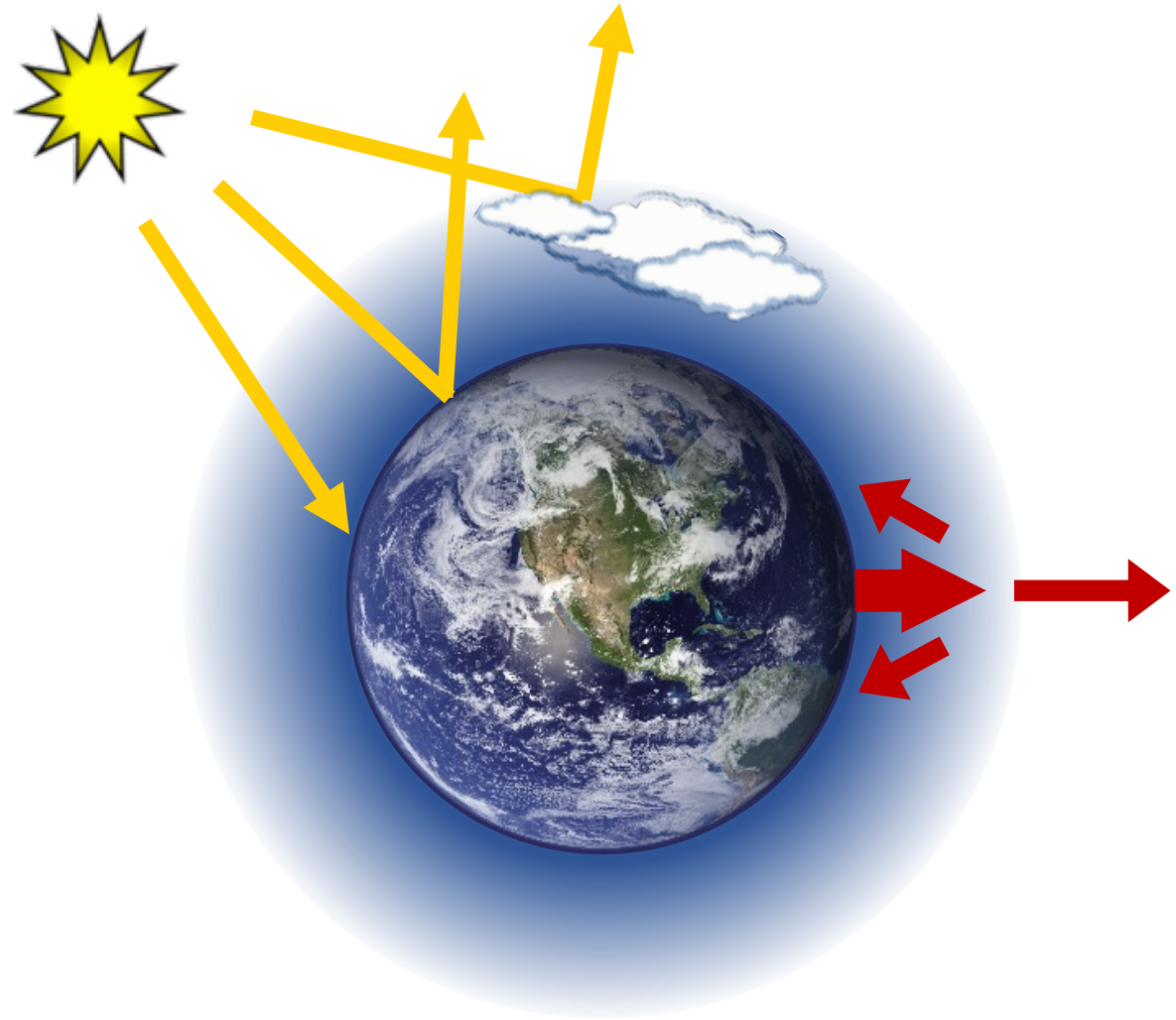
- Let's consider the case of GHG forcing
- At equilibrium:
energy in = energy out
- Immediately after increasing GHG concentration:
energy in > energy out
- This *instantaneous* radiative perturbation is called a **radiative forcing**



Perturbing radiative balance: GHG forcing

- Over time, the Earth will accumulate heat, causing warming
- Warming allows Earth to emit more radiation (Planck's law)
- This continues until “energy out” is large enough
- At the new equilibrium we again have

energy in = energy out

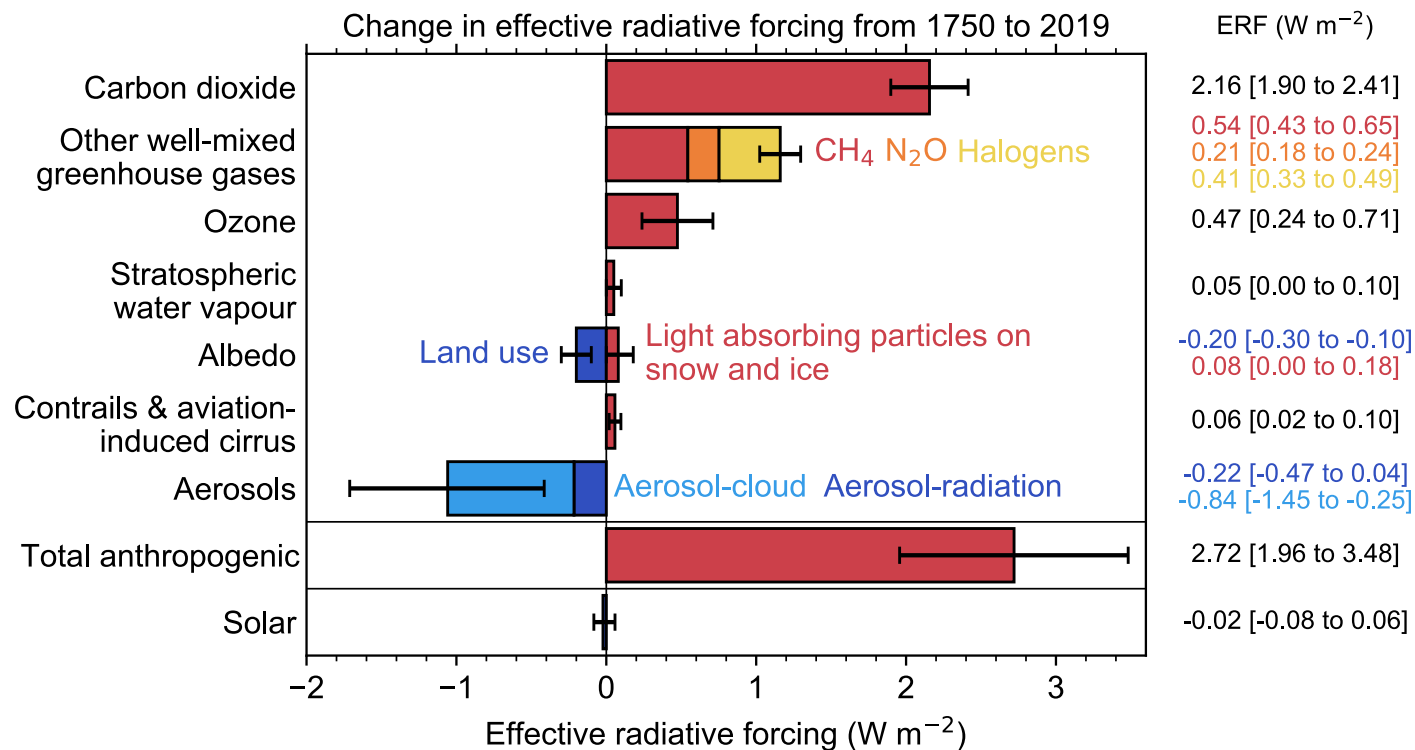


Types of radiative forcing

- Greenhouse gases are not the only type of radiative forcing
- Other examples of anthropogenic forcings:
 - Aerosol emissions (can reflect or absorb solar radiation)
 - Land use changes → surface albedo
- Examples of natural forcings:
 - Solar activity variations
 - Volcanic forcing (negative, because of dimming aerosols)

IPCC AR6 radiative forcings

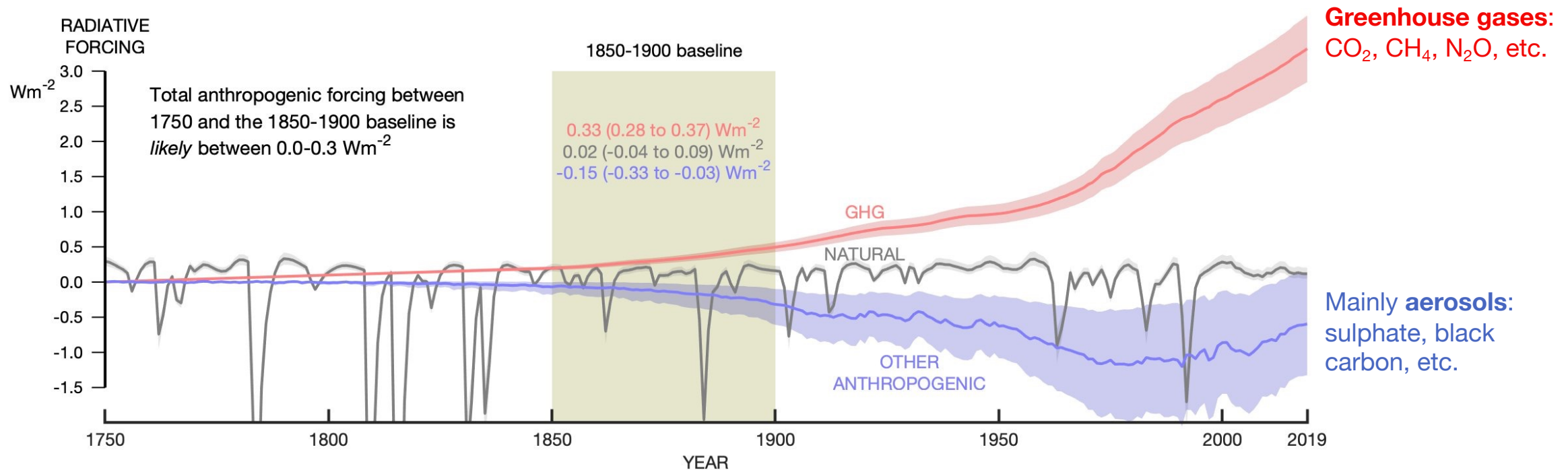
- Present-day forcings relative to 1750
- Main forcing is CO₂
- Methane and aerosol effects quite substantial too, but large uncertainties (error bars) for aerosols
- Volcanic forcing not included here (not relevant on long time scales)



IPCC AR6 WG1 Fig. 7.6

Aerosol forcing

- There are many types of aerosols. Most have a cooling effect (because they directly reflect sunlight, or because they promote the formation of white clouds)
- Aerosol forcing peaked in the late 20th century; now declining thanks to air quality regulations



Recap

- All bodies emit electromagnetic radiation; the amount of radiative energy emitted scales with temperature
- The natural greenhouse effect keeps us warm: without it global average temperature would be well below freezing
- Many atmospheric gases cause a greenhouse effect: the main ones are water vapour, carbon dioxide (CO₂) and methane (CH₄)
- Radiative forcing quantifies how a change in atmospheric or surface properties affects the Earth's radiation budget (in W/m²)

Further reading

- Global warming: Understanding the forecast (D. Archer), Part I
- IPCC AR6 WG1 Chapter 7, [The Earth's energy budget, climate feedbacks, and climate sensitivity](#)